Fitness Testing 1: Detailed Notes

Course: Personal Training Certification

Topic: Fitness testing, part 1 of 2

About these notes: Detailed notes from the Fitness Testing 1 lecture, written so the whole lecture can be studied without watching the recording. Every concept, example, analogy and class question is covered. For a short revision version see the Study Notes.

Abbreviations used in these notes

Short form	Meaning
PAR-Q	Physical Activity Readiness Questionnaire
NOC	No objection certificate (a doctor's written clearance)
BP	Blood pressure
SYS / DIA	Systolic / diastolic blood pressure (as shown on a BP monitor)
mmHg	Millimetres of mercury (Hg is the symbol for mercury), the unit of blood pressure
BW	Body weight
LBM	Lean body mass
BF	Body fat
VF / SF	Visceral fat / subcutaneous fat
EF	Essential fat
WC	Waist circumference
HC	Hip circumference
WHR	Waist to hip ratio
BMI	Body mass index
BIA	Bioelectrical impedance analysis
IMTG	Intramuscular triglyceride
TG	Triglyceride
RM	Repetition maximum (1 RM is the heaviest weight that can be lifted for one repetition)
FT	Fitness testing

Before this lecture

This lecture uses ideas from earlier topics. You should already be familiar with: the **five components of fitness** and their interdependence (cardiorespiratory endurance, muscular strength, muscular endurance, flexibility, body composition), the definition of **body composition**

and the make-up of body weight, the definitions of **muscular strength** and **muscular endurance**, **flexibility** versus **mobility** and the factors affecting mobility (tissue temperature), the **cardio lectures** (heart rate as the key measure in cardio) and **1 RM** from weight training.

Outline of the lecture

Part	Topic
1	Introduction: standardised tests, test norms, why test at all
2	Objectives of fitness testing (global and client)
3	Health screening before testing: PAR-Q and informed consent
4	The correct order of fitness testing, and why
5	Test 1: blood pressure
6	Body composition tests: overview and classification
7	2.1 Anthropometric data: waist circumference, waist to hip ratio, body mass index
8	2.2 Skinfold measurement (caliper)
9	2.3.1 BIA test
10	2.3.2 Bod Pod test
11	Wrap-up and preview of Fitness Testing 2

Part 1: Introduction

1.1 Standardised fitness tests

Fitness testing is taught in two parts, **Fitness Testing 1** and **Fitness Testing 2**. Together they cover the **standardised fitness tests** that are conducted worldwide in the same way, with the same **guidelines, procedures, methods and test norms**.

A **test norm** is a reference sheet. After a test is conducted, any outcome (for example a number or score) is checked against the test norm sheet, which shows where that person stands for the fitness component being tested.

Fitness is made of **five components**, so a person's fitness level cannot be assessed or declared "fit" from one or two components. Tests related to **all** the components must be conducted.

These tests were **not drafted by the course**. They were developed through research by international bodies working in fitness science and fitness research, and they are accepted and conducted worldwide.

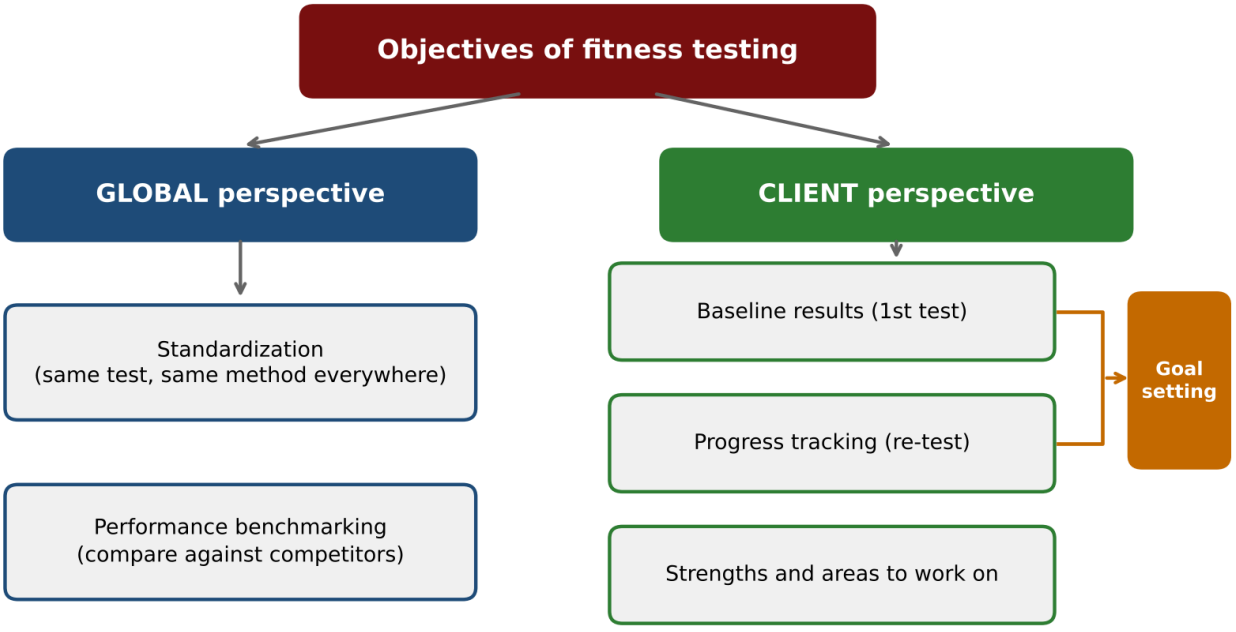
1.2 Why test at all?

Testing in general means **evaluating something to check its performance, quality or efficacy**. It exists in many fields:

- **Medicine:** doctors conduct tests to diagnose disease.
- **Manufacturing:** brands test a product before launch to see whether it will succeed.
- **IT:** software is tested heavily.
- **Education:** knowledge is tested in exams (the final exams of this course are an example).
- **Fitness:** a person's fitness is evaluated, with defined objectives.

Part 2: Objectives of Fitness Testing

The objectives are grouped into two perspectives: the **global perspective** and the **client perspective**.



Strengths and areas to work on also drive the customised workout plan

2.1 Global perspective

Standardisation. Because the tests use the same procedure and method everywhere, a given component is tested the same way all over the world. For example, muscular strength is not tested with a lat pull down in Mumbai, a shoulder press in Tokyo and a bench press in France. One of the worldwide tests of muscular strength is the **bench press**, and nobody may invent a different method. (The strength tests themselves are covered in Fitness Testing 2.)

Standardisation allows **comparison**: athletes or bodybuilders of one country can be compared with those of another on the same parameter.

Performance benchmarking. Suppose a country's athletes will take part in a global tournament. Unless it is known where the competition stands (for example South African athletes here, New Zealand there, the USA elsewhere), there is no milestone or benchmark to aim for. Benchmarking shows how far a team lags, or how far ahead it is. Being ahead is also useful information, because

then the standard has to be maintained. From a global point of view, fitness testing is all about performance benchmarking made possible by standardised methods.

2.2 Client perspective

If fitness testing helps the client, it also helps the trainer. A client who enjoys the benefits stays, renews training and becomes a **brand advocate** (tells others about the trainer).

1. Baseline results (the first test). The trainer knows nothing about a new client. Testing all five components gives the **baseline** that shows where the client stands today.

2. Progress tracking (the re-test). Only with baseline numbers can progress be tracked. After, say, 3 months of training a re-test shows how far the client has come. The trainer can show the client: "this is where you started and this is where we have brought you". This builds confidence and motivation, because numbers speak louder than words. It also shows that the trainer is **number-driven and goal-oriented**, not someone making vague promises such as "give me one year, I will do magic".

3. Goal setting. Baseline results (first test) and re-test results together support goal setting. Without baseline results goals would be set randomly. After a re-test the goals can be tweaked depending on whether the plan is working. Instead of vaguely saying "one year, two years", the trainer sets **short-term, mid-term and long-term goals** (actual milestones, for example at 3 months and 6 months). This keeps the client consistent and driven. When clients meet these goals they look up to the trainer and may ask for the bar to be set higher.

4. Strengths and areas to work on. The baseline shows what the client is good at and what needs work, which lets the trainer design a **customised and suitable workout plan**.

- **Wording matters.** Fitness is a service industry, so the word "weakness" is avoided. Use **strengths** and **areas to work on**. A client meeting the trainer for the first time may react badly to a list of "weaknesses".
- **"Strength" here does not mean only muscular strength.** It is the ordinary English word and can come from any of the five components.

Analogy: the debate competition. A school girl takes part in a debate and her teacher gives feedback afterwards. The weaknesses mentioned are pronunciation of certain words and grammar. The strength is good voice modulation (the line "what are the objectives of fitness testing" did not sound like a robot, which made it impactful). Strengths can lie in any field.

Using strengths. A client with very good cardiorespiratory endurance can be helped to maximise that potential and even build a career from it. A client with very good upper body muscular strength can be given more advanced upper body exercises and helped to nail those lifts.

Using areas to work on.

- A client with **poor flexibility** (flexibility has its own tests) gets a balanced plan in which not a single day passes without stretching: **mobility drills as well as static stretching**.
- A client with a **contraindication** can be protected. A contraindication is any movement that causes discomfort or pain (the word is often used by physiotherapists). It may be in the upper limb, lower limb or back, for example raising the hand with the elbow bent. If a particular machine or exercise bothers the client, it can hinder performance, so an alternative is chosen. This avoids the risk of injury.

Fitness testing is therefore a powerful tool for designing the most suitable workout plan.

Part 3: Health Screening Before Fitness Testing

3.1 What is screening?

Before the correct order of tests is discussed, there is a **pre-step** that takes place even before fitness testing: **health screening**.

The class was asked what "screening" means. Answers included checking records, a health check and filtering. The simplest way to understand it is **baggage screening at airports, metro stations and hotels**, or X-ray screening. Screening means **scanning** something. Security personnel scan everybody's luggage so that people in a public area are safe and nothing dangerous is found. In the same way, health screening makes sure that the person whose fitness test is about to be conducted is **safe and sound** and that no problem will occur during the test.

The course slides define it as: *a procedure performed on asymptomatic populations to assess the likelihood of a particular disease.*

Health screening does **not** require inviting a doctor to the gym. It uses **two simple forms**. The trainer reads the form to the client (or the client reads it), the client fills it in and signs it. The two forms are the **PAR-Q** and the **informed consent form**. Sample copies of both are in the course slides and can be printed and used later, for example if a gym has no such process (introducing one is usually welcomed). The forms are also a requirement of the international registration body, so trainers must be aware of them.

3.2 PAR-Q

PAR-Q stands for **Physical Activity Readiness Questionnaire**. It is a set of about 10 questions (the sample in the slides has 10) about the client's **medical background**: for example heart-related conditions, joint-related conditions and neurological disorders such as epilepsy.

The questions are **close-ended**: next to each there are only two options, **Yes or No**, and the client selects one. There are no lines for writing an essay about a past problem.

Clearance rule. The PAR-Q is considered cleared, and the fitness test can be conducted, only when **all answers are No**. A single **Yes** is a signal to stop and ask the client about it. To be sure the client is still eligible for fitness testing, the client must bring a **NOC (no objection certificate) from their doctor**.

Why documentation matters. Today a client may blame the trainer and threaten legal action very quickly. Getting everything documented and signed before the journey starts means nobody can hold the trainer responsible later. The more detailed the form, the better.

The sample PAR-Q in the slides (ages 15 to 69) contains these questions, each answered Yes or No:

No.	Question (summary)
1	Has your doctor ever said you have a heart condition and that you should only do physical activity recommended by a doctor?
2	Is there any history of heart disease in your family?
3	Do you feel pain in your chest when you do physical activity or exercise?
4	In the past month, have you had chest pain when you were not doing physical activity?
5	Do you lose your balance because of dizziness, or do you ever lose consciousness?
6	Do you have a bone or joint problem (for example back, knee or hip) that could be made worse by a change in your physical activity? (If yes, give details.)
7	Do you suffer from asthma, diabetes, epilepsy or high blood pressure? (If yes, give details.)
8	Do you have any other medical or physical condition (such as diabetes, cancer, osteoporosis)?
9	Do you have any current injuries or conditions, and if so, are they being treated by a doctor or other health professional such as a physiotherapist?
10	Do you know of any other reason why you should not do physical activity or exercise?

The form ends with a note: if you answered YES to any question, check with a member of staff before taking part, because you may need to be referred to your doctor first.

3.3 Informed consent

The word **informed** means there is a **script** that the client reads, or that the trainer reads aloud. It informs the client about:

- the kind of **machines** that will be used,
- the **exercises** and activities (strength training, cardio, stretching and so on),
- the **possible risks** involved.

This form is not only for fitness testing. It covers the **overall fitness journey**, because a client who has signed up is tested first and then the journey begins.

Consent is the closing statement in which the client says they are fully aware of everything and **willingly agree** to take part in the fitness journey or fitness test, followed by the **signature**. It is a win-win: the client is aware and the trainer is protected.

The sample consent form in the slides includes statements that the programme includes exercises to build the cardiorespiratory system (heart and lungs) and the musculoskeletal system (muscles, joints and bones) and to improve body composition. Exercise may include treadmill, running, walking, bike, rowing, group aerobic exercises, weight training and exercises for mobility and flexibility. The client confirms that it is their responsibility to inform the trainer of any health problems, injuries or pregnancy, to obtain written medical clearance if required, to monitor their own condition and stop and inform the trainer if unusual symptoms occur, and that they have read the form and their questions have been answered.

Student question: "In every gym, consent forms are mandatory, aren't they?"

A: If a gym already has its own format, follow it. Some gyms do not like changes to their standard process, and there is no need to go against the system. Where nothing exists, a trainer can

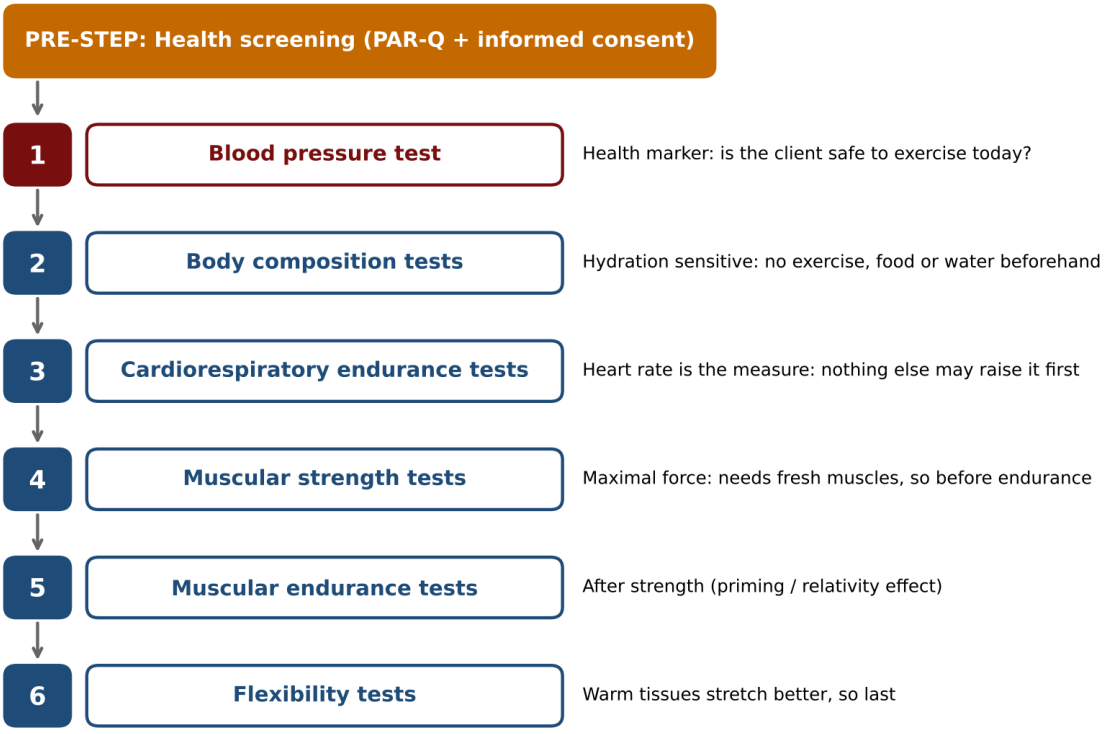
introduce one. It will be seen as a positive step, working for the organisation and not only for oneself.

Once these two formalities are done, fitness testing can begin.

Part 4: The Correct Order of Fitness Testing

4.1 Why order matters

There are five components and several tests for each. The question is which component's tests come first and which last, so that the results are **valid and reliable**.



Steps 2 to 6 are the five components of fitness. The order is for accurate results, not a ranking of importance.

Step	Test	Why it sits here
Pre-step	Health screening (PAR-Q and informed consent)	Safety and legal protection before anything else
1	Blood pressure test	Health marker for the day of testing
2	Body composition tests	Needs a neutral, unchanged body state
3	Cardiorespiratory endurance tests	Heart rate must rise only because of the cardio task
4	Muscular strength tests	Maximal force needs fresh muscles
5	Muscular endurance tests	Easier after the strength test (priming effect)
6	Flexibility tests	Warm tissues lengthen better

Steps 2 to 6 are the **five components of fitness**. Step 1 is an additional **health marker** test. The numbers 1 to 6 are an **order**, not a rank: blood pressure is not "most important" and flexibility is not "least important". All components are equally important (see the interdependence of components in the Intro to Fitness lecture). The order simply gives the most accurate results for all tests.

4.2 Step 1: blood pressure comes first

The PAR-Q and consent form describe the client's medical background over the last 3 or 6 months. The client's condition **on the day of testing** (especially the very first time) also has to be checked, because nothing is known about the client. A **blood pressure test** shows whether the client is ready to be taken through the tests today.

If BP is already elevated or high and the client has not mentioned any such history, physical activity during the tests could push it up further, and things could go out of control. It must therefore be normal at the start.

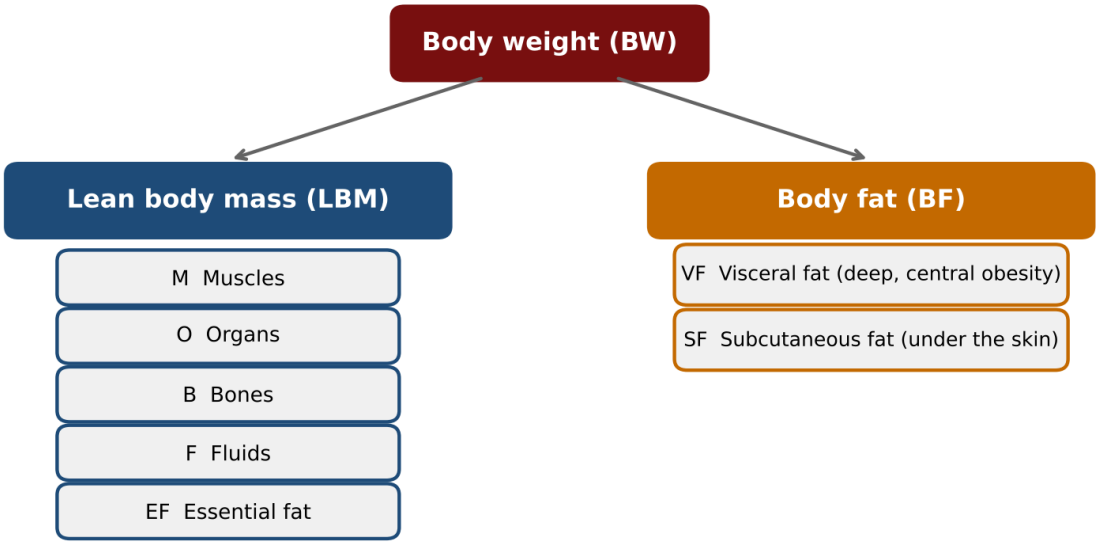
If BP is elevated or high on the day:

1. Make the client sit down in a calm, relaxed environment and offer water.
2. Check again after some time. If it is back to normal, begin.
3. If it stays high, **avoid the fitness test** that day. Try on the second day and the third day.
4. If BP is high on all three days, it can indicate a high blood pressure condition, and the client should be told to see a doctor if they have not already.

Three days are used because one high reading may be due to a stressful day, physical activity before coming to the gym, or some other factor. The aim is to be certain that the day on which testing is done is a normal day.

4.3 Step 2: body composition second

Body composition is defined as **the ratio of body fat and lean body mass** (answered by the class). Body weight can be divided into two main parts:



Body composition = the ratio of lean body mass to body fat

Lean body mass (LBM): Muscles, **O**rgans, **B**ones, **F**luids and **E**ssential **F**at (memory aid M, O, B, F, EF). **Body fat (BF): visceral fat (VF)** and **subcutaneous fat (SF)**.

Why first? Body weight changes before and after a workout, because physical activity causes metabolic changes: sweating causes **fluid loss** and **salt depletion**. To test body composition in the most **accurate and reliable** way, the test must be done when everything is **neutral** and nothing in the body has changed. That is why, for the popular BIA machine, people are told to do the test **first thing in the morning before eating or drinking anything**. Body composition tests are therefore conducted before any other test.

4.4 Step 3: cardiorespiratory endurance third

In the cardio lectures the key factor was the **heart rate** (just as RM is the key factor in weight training). In both cardiorespiratory endurance tests, heart rate is monitored.

Reasoning. Suppose a client does 10 body weight squats, 10 push-ups and then 10 minutes of jogging, and the heart rate is then found to be 132 (a hypothetical number). There is no way to say how much of that heart rate comes from the jogging, because squats and push-ups also raise it. The increase should be **solely the contribution of the cardio task**. Muscular strength and muscular endurance tests would raise heart rate, and so would **flexibility** (stretching). So the cardiorespiratory endurance test is done before all of them, to get the most accurate result.

4.5 Steps 4 and 5: strength before endurance (class vote)

The class was asked to vote on whether muscular strength or muscular endurance comes first. The majority said strength first, which is **correct**. (When this vote is held, many people say endurance first.)

Definitions recalled.

- **Muscular strength:** the ability of the working skeletal muscles to contract and generate **maximum force in one single effort**.

- **Muscular endurance:** the ability of the working skeletal muscles to contract **submaximally for a long period of time without getting fatigued**.

Reason 1: fatigue. Take the pectoralis major. A **bench press** (a 1 RM style test) tests strength and **push-ups** test endurance. If push-ups are done first, the muscle is fatigued, and it would then be expected to generate maximum force, which is unfair. The muscle must be fresh so that it can contract and generate maximal force.

Reason 2: the priming or relativity effect (it helps endurance too). Imagine sitting in a car at 100 km/h, which feels very fast on Indian roads. Later you sit in a car (say a Lamborghini) at 200 km/h, double the speed. When that car comes back down to 100 km/h, 100 km/h no longer feels as fast, because you are coming down from 200. In the same way, a muscle that has gone through a **1 RM** test (maximum force in a single effort, the "200 km/h" test) finds the endurance test (the "100 km/h" push-ups) **less challenging**. A client who could do X push-ups when push-ups come first might do more than X when push-ups follow the strength test. So both tests benefit from this order.

4.6 Step 6: flexibility last

Flexibility and mobility go hand in hand, and the factors that affect mobility affect flexibility too. One of those factors is **tissue temperature** (from the stretching lecture): warmer tissues extend better, and colder tissues are stiffer.

Flexibility is the ability of the soft tissues to **lengthen passively**, so flexibility tests involve stretching the muscles. If flexibility were tested at position 1, 2 or 3, no physical activity would have been done yet and the tissues would be cold. After steps 3, 4 and 5 the tissues are warm and lengthen better. So flexibility is tested **last**, to expect the best possible results.

Analogy: making tea. If a host puts ginger and cardamom in the utensil first, roasts them, then adds tea, and only at the end adds water and milk, the tea will not turn out right (guests would end up ordering tea from outside). Even a simple routine activity needs the correct order for a successful result, and so does fitness testing.

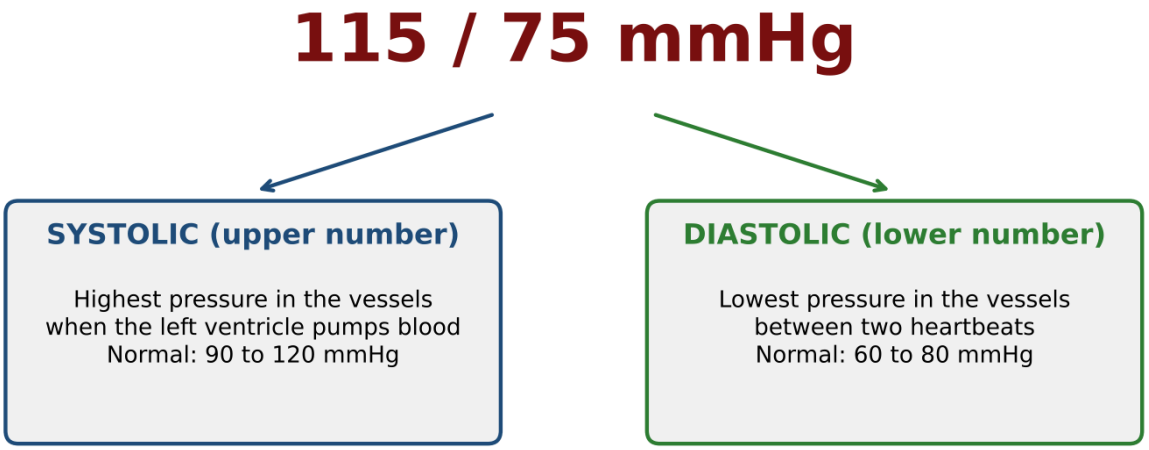
Student question: Do we have to test mobility too?

A: Flexibility and mobility go hand in hand. **Mobility** is the ability of joints to move **freely and actively through a full range of motion without discomfort**. Some flexibility tests check the range of motion at a joint, so mobility is tested side by side. It is not listed separately only because there is no component named mobility; the tests are named after flexibility.

Part 5: Test 1, Blood Pressure

5.1 Reading a BP report

A BP reading looks like **115/75 mmHg**. In everyday language these are called the "upper one" and the "lower one". The professional terms are:



mmHg = millimetres of mercury (Hg is the symbol for mercury)

- **Systolic BP** (the upper, larger number): when the heart beats and the **left ventricle pumps blood** into the vessels, the pressure in the vessels is highest. This is the systolic pressure.
- **Diastolic BP** (the lower, smaller number): in the gap between two heartbeats the pressure falls. The lowest pressure in the vessels is the diastolic pressure.

Normal ranges: systolic **90 to 120 mmHg** and diastolic **60 to 80 mmHg**. **mmHg** is the unit: millimetres of mercury (Hg is the symbol for mercury). A reading of 115/75 lies in both normal ranges, so it is **normal BP**. (Many animation videos show systolic and diastolic clearly.)

5.2 The measuring device

Older manual equipment, still used by some nurses and doctors, has a balloon that is pumped, a tube connected to a cuff that inflates, and a rectangular meter containing mercury that rises and falls (a **sphygmomanometer**). The person testing also uses a stethoscope to hear the pulse sounds.

No such skill is needed here. A **simple automatic digital machine** is used (for example the Omron brand, which is quite reliable). The tube from the cuff is plugged into the monitor, the cuff is wrapped on the upper arm, and the start button is pressed. The machine does everything. The display shows **SYS** (systolic), **DIA** (diastolic) and also the **pulse rate** per minute. After pressing start a mild sound and vibration indicate that the cuff has started inflating by itself.

5.3 Seven-step procedure

Step	Instruction
1	Plug the tube end into the device
2	Tie the cuff on the upper arm with both the tube and the arm (palm) facing up
3	Position the cuff edge **2 to 3 cm above the elbow** so that it does not cover the elbow joint
4	The cuff should be **snug but not too tight**: two fingers should fit under it
5	The tube should pass along the **medial side** of the arm (not anterior, not lateral)
6	The cuff should be at the **same height as the heart** (use a table or armrest)
7	Correct posture: legs not crossed, feet flat on the floor, back supported; **no talking** and no stress, by the client and by the tester

5.4 Blood pressure classification

After the numbers are ready they are checked against a chart, like the test norm sheets for most upcoming tests.

Blood pressure category	Systolic mmHg (upper number)		Diastolic mmHg (lower number)
Low blood pressure (hypotension)	Less than 90	Or	Less than 60
Normal	90 to 120	And	60 to 80
Elevated	120 to 129	And	Less than 80
High blood pressure (hypertension stage 1)	130 to 139	Or	80 to 89
High blood pressure (hypertension stage 2)	140 or higher	Or	90 or higher
Hypertensive crisis (consult a doctor immediately)	180 or higher	And/or	Higher than 120

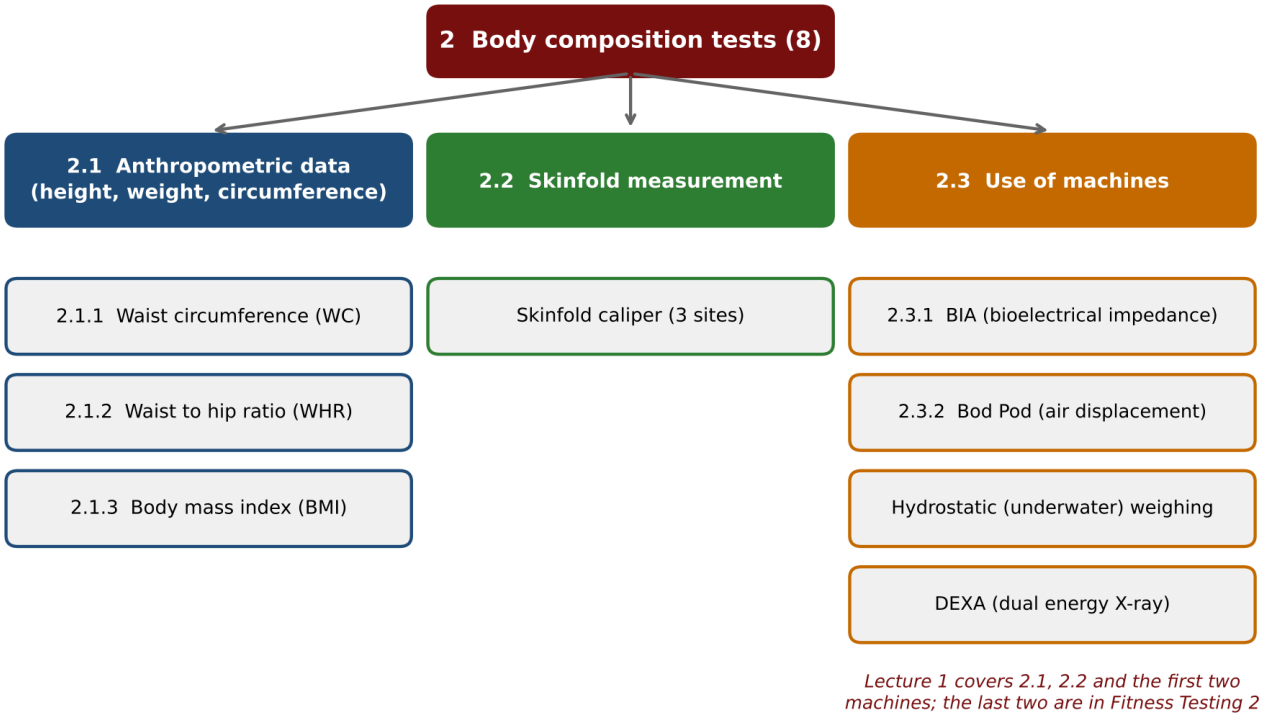
Language. "Hypo" means less (hypotension is low BP), "hyper" means excessive (hypertension is high BP).

Applying the chart. 115/75 falls in both normal ranges, so it is normal. If **either** the systolic is below 90 **or** the diastolic is below 60, it is low BP. At 140 or higher (systolic) or 90 or higher (diastolic) it is high blood pressure stage 2.

Safety link. If the client already had hypertension stage 1 and a physical activity pushed BP further to stage 2 or a hypertensive crisis, it could be very risky. This is why BP is a **health marker** checked first.

Part 6: Body Composition Tests, Overview

Body composition has **eight tests** that are grouped into **three categories**. This makes it the **longest** component of fitness testing (it may be started in one lecture and finished in the next).



Lecture 1 covers 2.1, 2.2 and the first two machines; the last two are in Fitness Testing 2

Category	Tests	Machine needed?
2.1 Anthropometric data	Waist circumference (WC), waist to hip ratio (WHR), body mass index (BMI)	No
2.2 Skinfold measurement	Skinfold caliper test (3 sites)	No (a small hand device)
2.3 Use of machines	BIA, hydrostatic (underwater) weighing, Bod Pod (air displacement), DEXA (dual energy X-ray absorptiometry)	Yes

The goal is to be **aware of all eight tests**, not to conduct all of them. After the pros and cons of each have been discussed (finishing in Fitness Testing 2), the most suitable test will be chosen.

Numbering system. The number 2 stands for the second type of fitness test (body composition). Under it, 2.1 is the first category (anthropometric data), 2.1.1 to 2.1.3 are its tests, 2.2 is skinfold measurement and 2.3 is use of machines (2.3.1 BIA, 2.3.2 Bod Pod). This makes it easy to remember which test belongs to which category.

Pattern followed for each test: objective, procedure (site or steps), norms or results, then advantages and disadvantages.

Part 7: 2.1 Anthropometric Data

7.1 What is anthropometry?

Anthropo means the **human body** and **metric** refers to **measurements and proportions**. **Anthropometry** is the scientific study of the measurements and proportions of the human body, and **anthropometric data** are the numbers collected in this study.

Three key measurements are taken:

1. **Height**, measured with a **stadiometer**. The person stands bare feet on a small platform connected to a pole, with the hips, shoulder blades and head aligned to the pole. A headboard is lowered to touch the **crown of the head**. The pole carries a scale in feet, inches and centimetres, from which the reading is taken.
2. **Weight**, measured with a weighing scale.
3. **Circumference**, the length of the boundary of something, measured with a tape (the same kind a tailor uses for chest, shoulder and lengths).

Slide guidelines (height and weight).

- *Height:* explain the procedure, remove shoes, stand tall and straight with shoulders relaxed, hands at the sides, thighs together and feet flat and slightly apart. Heels, buttocks and shoulder blades should touch the surface of the measuring board. Lower the headboard until it firmly touches the crown of the head. Record height to the nearest quarter inch or cm, rounding up or down.
- *Weight:* explain the procedure, remove shoes, stand in the centre of the scale, stand still with even weight on both feet and arms hanging freely at the sides, and record the weight shown.

Some tests use only circumference, others use height and weight. The three tests of this category are WC, WHR and BMI.

7.2 Objectives of the three tests

Test	Objective
2.1.1 Waist circumference (WC)	Tells whether there is a **health risk due to visceral fat** (central obesity)
2.1.2 Waist to hip ratio (WHR)	Tells whether there is an **overall health risk**
2.1.3 Body mass index (BMI)	Places the person in a category: **underweight, normal weight, overweight or obese**

Visceral fat is the deep fat towards the centre, away from the surface, which is why it is also called **central obesity**. **Subcutaneous fat** lies towards the skin surface.

Important: WC only says whether the **risk** is high or low. It does **not** give a visceral fat percentage. WHR is not specific to visceral fat; it gives the general overall health risk.

7.3 2.1.1 Waist circumference (WC test)

Site. Place the measuring tape **across the iliac crest** (the highest point of the ilium, which can be felt by standing and touching it). Across the iliac crest the tape also passes through the **belly button**, and this is the **widest part of the waist**.

Touch and consent. A test with physical touch needs **permission**, whatever the gender. If the client is comfortable, the trainer takes the measurement. If not, the trainer shows the position on their own body and the client places the tape on themselves, and the trainer notes the reading on the tape.

Language with clients. Avoid jargon. A client told "put it across your iliac crest" will ask "where?". Say instead **"the widest part of your waist"**, which automatically falls across the iliac crest.

Normal values (for Indians, and applicable to Asians overall). These differ for males and females, and are defined as **low health risk**.

	Low health risk (normal)
Males	WC less than **35.5 inches (90 cm)**
Females	WC less than **31.5 inches (80 cm)**

The tape has inches on one side and centimetres on the other. These values were not set by the course. They were tried, tested and suggested by research bodies and international bodies.

The slides summarise it as: measures risk due to central obesity or visceral fat (risk for lifestyle disorders), with the same Asian values, and a diagram of the iliac crest, visceral fat, abdominal muscles and subcutaneous fat. Their guidelines say to demonstrate to the subject, have them stand upright and relaxed, wind the tape around the greatest anterior extension of the abdomen (usually at the level of the umbilicus and directly above the iliac crest), ask whether the subject wants to be measured or measure themselves, and record the measurement.

7.4 2.1.2 Waist to hip ratio (WHR test)

Formula. The ratio of waist circumference to hip circumference:

WHR = WC / HC

Both measurements must be in the **same unit** (inches with inches, or cm with cm; never mix them).

Sites. The waist site here is **different** from the WC test above, even though the same letters "WC" are used.

Measurement	**Site**	**Plain wording for clients**
Waist (WC) for WHR	The **narrowest part of the waist**: the midpoint between the last rib and the iliac crest	"Narrow part of the waist"
Hip (HC)	The **widest hip area** (the slides: maximum circumference of the buttocks)	"Widest part of the hips"

There is no narrow or wide choice for the hips and no special medical term, just the widest hip area.

Normal values (low health risk).

	WHR
Males	Less than **0.95**
Females	Less than **0.86**

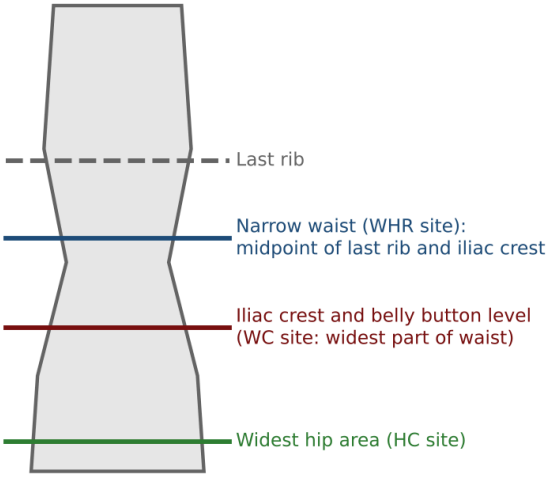
Worked case study. A male client's narrowest waist is 36 inches and his hip is 40 inches (inches for both).

WHR = 36 / 40 = **0.90**

For a male the limit is 0.95, so 0.90 is below it: **low overall health risk**.

Using the **same numbers for a female** (the slides use a female for this case): the limit is 0.86 and 0.90 is bigger than 0.86, so the result is **high overall health risk**. The same number gives different results for the two sexes.

Measuring sites (schematic, not to scale)



Test	Male	Female
WC (low risk)	< 35.5 in (90 cm)	< 31.5 in (80 cm)
WHR (low risk)	< 0.95	< 0.86

WHR = waist circumference / hip circumference
(use the same unit for both)

Case: waist 36 in, hip 40 in
WHR = 36 / 40 = 0.90
Male: 0.90 < 0.95 -> low overall risk
Female: 0.90 > 0.86 -> high overall risk

7.5 2.1.3 Body mass index (BMI test)

The first two tests use only circumference. BMI uses the other two anthropometric measurements, **weight and height**, through a formula.

Formula.

BMI = body weight (kg) / [height (m)]²

The unit of BMI is **kg/m²**. Squaring means multiplying a number by itself once ($2^2 = 2 \times 2$, $3^2 = 3 \times 3 = 9$), so the height in metres is multiplied by itself. Online BMI calculators exist, but a trainer should know the formula and calculation (for example when asked in an interview).

Converting height to metres. Heights usually come in a mixed form such as 5 ft 8 in.

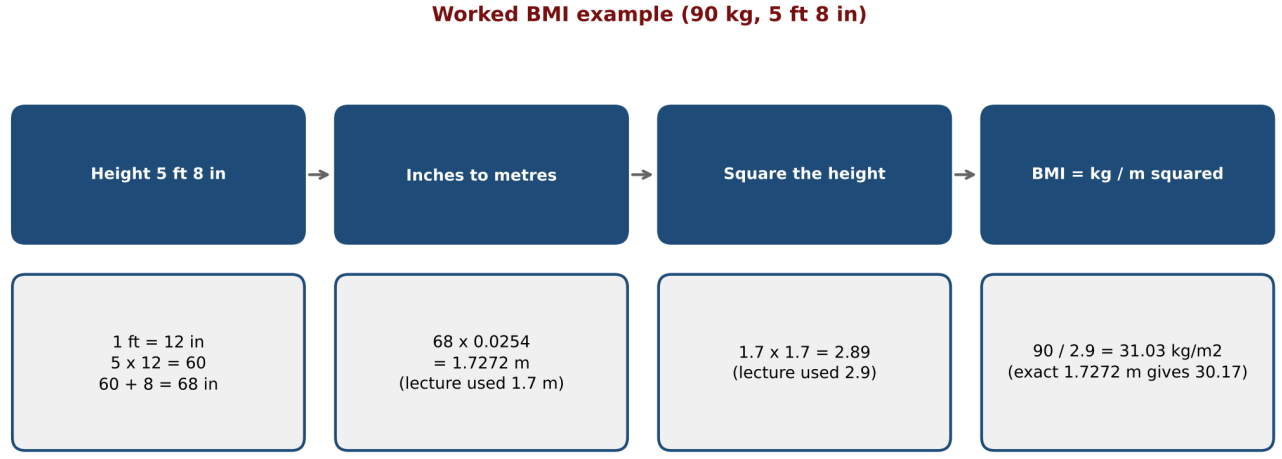
1. Make it a single unit (inches): 1 ft = 12 in, so 5 ft = 5 × 12 = 60 in. Then 60 + 8 = **68 in**.

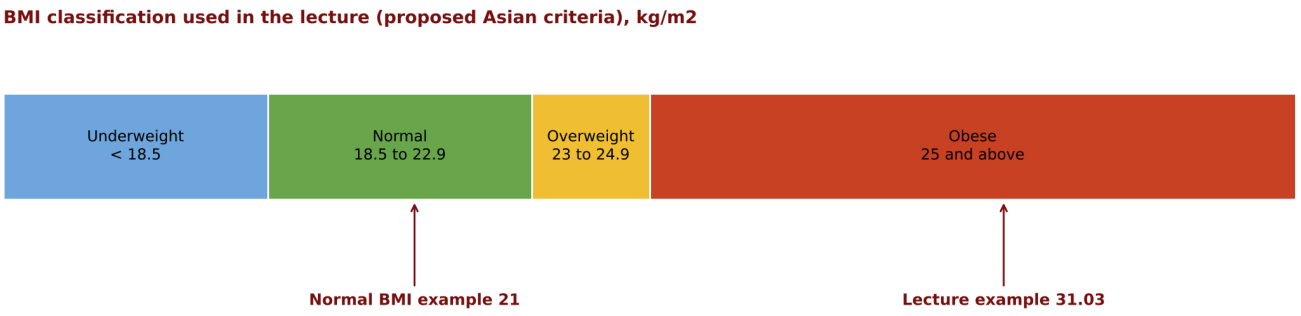
2. Multiply the inches by the constant **0.0254** to get metres: 68 × 0.0254 = 1.7272 m.

Worked example (as done on the board). Body weight 90 kg, height 5 ft 8 in. The height was taken as **1.7 m**, then 1.7 × 1.7 = 2.89, written as **2.9**:

BMI = 90 / 2.9 = **31.03 kg/m²** (written 31.02 and 31.03 on the board)

(Using the exact 1.7272 m gives about 30.17. See the points to confirm; the category is the same either way.)





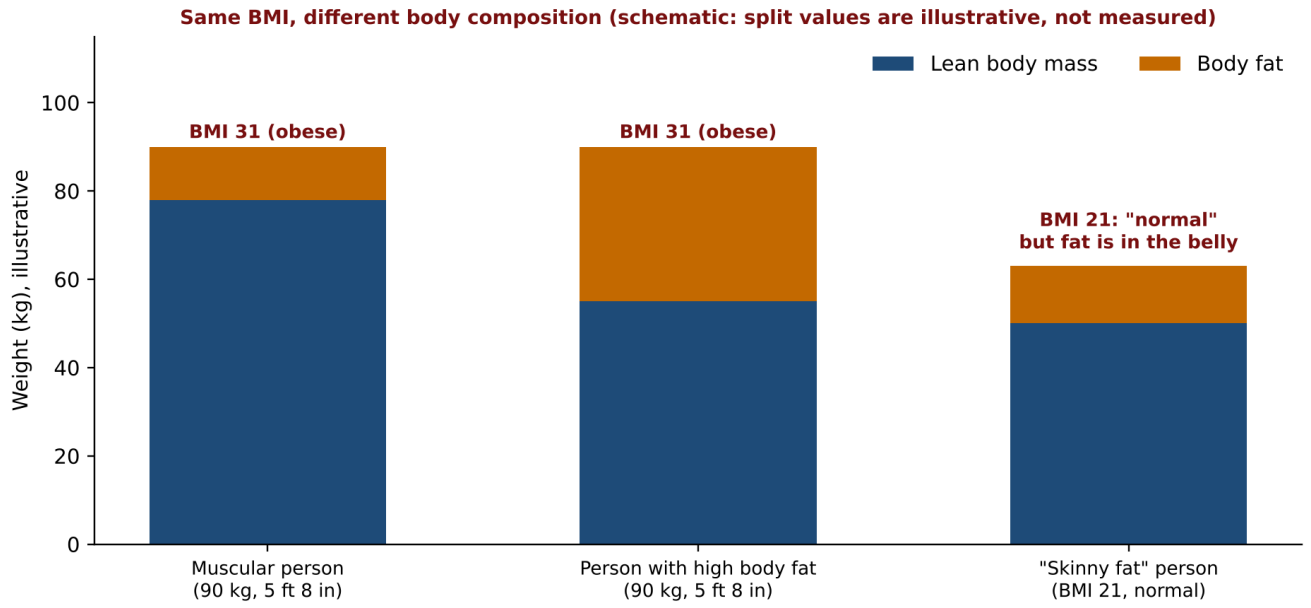
BMI classification (proposed Asian criteria, as in the slides). The slide chart is adapted from The Lancet 2004;363:157-63.

Category	BMI (kg/m²)
Underweight	Less than 18.5
Normal range	18.5 to 22.9
Overweight	23 to 24.9
Obese	25 and above (the slide prints "> 25")

At 31, the example client is placed in **obese**.

Shortcomings of BMI

The lecture stressed that BMI should not be relied on as accurate, because of loopholes:



1. It does not differentiate body fat from lean body mass. Suppose a client's details (90 kg and 5 ft 8 in) arrive on WhatsApp or from the gym manager before the first meeting, and the trainer calculates BMI as homework and concludes that he is obese. At the first meeting the client turns out to be a **muscular person**. The trainer would feel awkward saying "I thought you were obese as per your BMI". Two people of the same height and weight (the slide shows a bodybuilder and an obese person on the same scale, both 90 kg) can have very different compositions, because **weight comprises both lean body mass and body fat**. People with the same weight cannot be assumed to have the same body composition.

2. "Skinny fat". Suppose the data gave a BMI of 21, which is **normal**. The client may actually be skinny fat: the upper and lower limbs are lean (no visible fat) but there is **adiposity in the belly region**. Calling this person normal is also incorrect, because BMI does not count body fat; it only looks at weight and height.

What the three anthropometric tests cannot do

None of the three tests gives an actual **body fat percentage**. The ideal body fat levels (from the body composition lecture and the slides) are **less than 15% for males** and **less than 20% for females**, so a percentage number is needed to judge ideal body composition. The slides illustrate these levels with figures from 45% down to 15% for females and 35% down to 8% for males.

The three tests are **not discarded or degraded**. They are used, but **only as short-term or mid-term goals** for the client (for example: "as per the WC test you are 2 inches above the normal value, so your short-term goal is to get below it so there is no health risk"). They cannot be relied on for body composition. The search for a percentage continues with the next tests.

Part 8: 2.2 Skinfold Measurement

8.1 The test

There is only **one test** in this category: the **skinfold test**, using a **skinfold caliper**. The caliper looks like the compass used to draw circles: it has two arms with a pointed surface at the end of each. The trainer pinches and holds the skin to make a **skin fold** and uses the caliper to **measure the thickness** of that fold.

The slides show the steps: pinch the skinfold between the fingers, then place the caliper on the skinfold to measure the fat percentage. Calipers come in several designs (four types are shown), but the process is the same.

What it measures and the principle. A pinch of skin and fat measures the **subcutaneous fat** (not the deep, internal fat). The principle: **the higher the subcutaneous fat, the higher the body fat percentage.**

8.2 Sites (3 sites each for women and men)

	Women	Men
Site 1	**Triceps fold**	**Chest:** midpoint between the top of the armpit and the nipple
Site 2	**Suprailiac fold:** supra means superior, iliac refers to the iliac crest, so the fold just above the iliac crest on the waistline	**Abdomen:** 1 inch from the belly button (left or right, anywhere)
Site 3	**Mid thigh**	**Mid thigh**

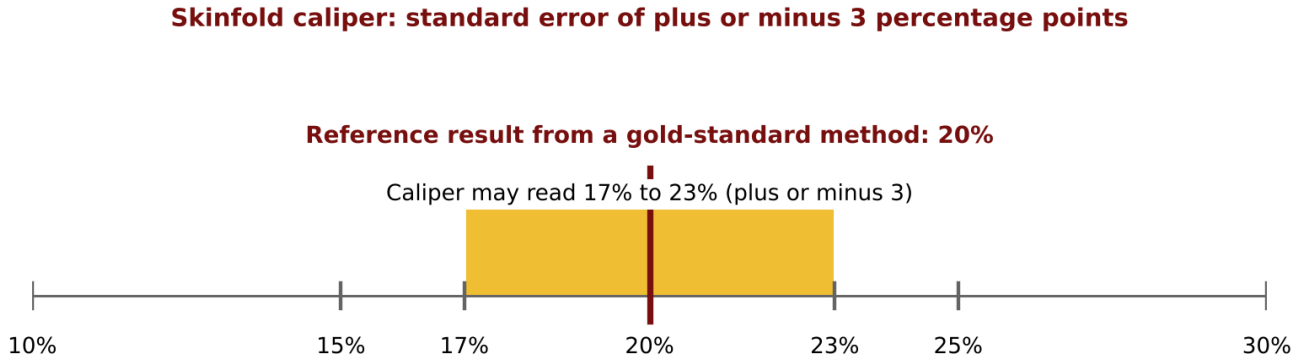
The mid thigh is the one site common to both sexes.

8.3 Advantages and disadvantages

Advantages	Disadvantages
Portable: small and handy, fits in a bag or a deep jacket pocket	**Measures only subcutaneous fat** (no visceral fat), so it does not tell how much is subcutaneous and how much is visceral; it cannot be relied on alone
Easy to use: once the three sites for each sex are known, there is no rocket science	**High standard error of plus or minus 3%**
Time saving: it takes only a few seconds	**Not suitable for the very obese or the very lean**
	Pinching discomfort: not everyone is comfortable being pinched

Standard error explained. Calipers range in price from about Rs 300 to Rs 5,000, and more spending may give better accuracy, but the method itself is still not considered fully accurate, even for an expensive one, because accuracy comes from the **type of test** and the steps taken, not from

the price. Accuracy of any test is compared with the **gold standard**. If the gold standard says a person has **20%** body fat, the caliper result can fall anywhere from **17% to 23%** (plus or minus 3 of that number).



Why not suitable for extremes. A very lean person may have so little fat that no fold can be pinched, so nothing can be measured. A very obese person may have more subcutaneous fat than the maximum opening of the caliper arms.

The slides list the same points: easy to use once the skill is mastered, does not require much time, easy portability; error of plus or minus 3% (compared with underwater weighing), does not differentiate subcutaneous from visceral fat, not ideal for the obese and very lean, and individual preference about being pinched.

From this test onwards, the advantages and disadvantages of each test are noted so that a final choice can be made.

Part 9: 2.3 Use of Machines: the BIA Test (2.3.1)

9.1 What BIA is

The **BIA machine** is the machine seen in most gyms that looks like a weighing scale: people stand on it and hold a handle connected by a chain or rope at chest level. It is often wrongly called the "BMI machine". It is called that because one of its results is BMI, which it calculates automatically. But the correct name is the **BIA machine**.

BIA = Bioelectrical Impedance Analysis.

Part	Meaning
Bio	Life, a living organism
Electric	Electric current
Impedance	Resistance
Analysis	Study

The slides describe it as a test that measures the difficulty level of an electric current moving through the body (resistance to the passage of current in the body).

9.2 Principle

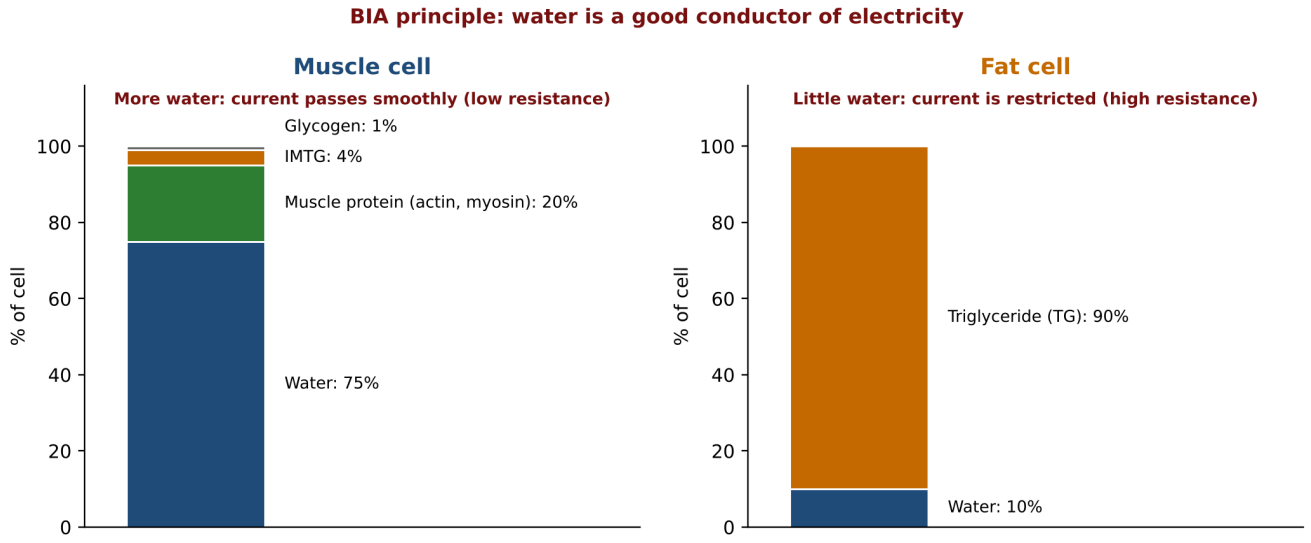
A person stands barefoot. The platform has **metal plates** (in some machines the whole platform is metallic, in others a plastic platform has a metal footstep). These are **electrodes** through which a very mild current enters the body. It is not felt as anything more than a mild vibration. The machine analyses **how much difficulty the current faces moving through the body**: is it resisted and restricted, or does it move smoothly? From this it works out who has more fat and who has more muscle.

Key scientific principle: water is a good conductor of electricity. Water can amplify an electric current (the reason one is told to stay away from open wires near water).

Comparing muscle and fat. From the body composition lecture:

	Muscle cell	Fat cell
Water	**75%**	**10%**
Other content	20% muscle protein (actin, myosin), 4% intramuscular triglyceride (IMTG), 1% glycogen (total 100%)	90% triglyceride (TG)
Effect on the current	Plenty of water, so the current **passes smoothly**	Little water, so the current is **restricted**

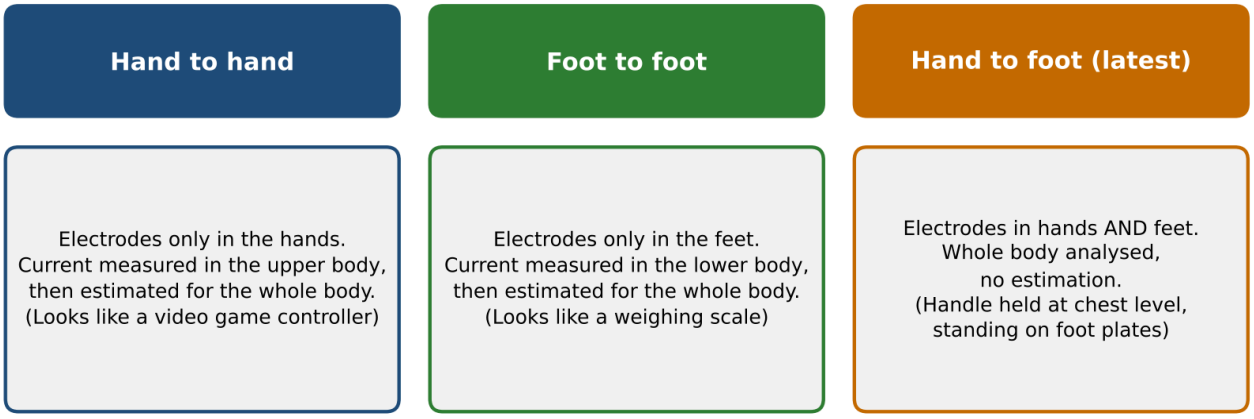
A person with **more fat mass** has more fat cells, hence a smaller proportion of water per cell, so the current faces restriction. A person with **more muscle mass** has more water in the body, so the current passes smoothly. The machine reads the resistance level and then reports fat mass and muscle mass percentages.



The slides present the same idea: lean body mass has higher water and salts and lesser fat (it conducts, allowing the current to pass faster), while adipose tissue has low water and higher fat content (it insulates, so the current passes slowly).

Compared with earlier tests. The anthropometric tests gave no percentage. Skinfold gave a percentage that was not fully reliable because it ignores visceral fat. BIA considers **both visceral and subcutaneous** fat and gives a percentage.

9.3 Types of BIA machines



- **Foot to foot:** looks like a weighing scale. Current enters through the feet only, the machine analyses the **lower body** and estimates for the full body.
- **Hand to hand:** looks like a video game controller. Electrodes are only in the hands, so current enters only through the hands, the machine analyses the **upper body** and estimates for the full body.
- **Hand to foot (latest machines):** electrodes in the hands **and** the feet, so there is **no estimation** and the entire body is analysed.

Price and features. BIA machines come in a wide price range, from about **Rs 10,000 up to Rs 45 lakh** per machine. High-end machines give **segmental analysis**: fat mass and muscle mass for the arms, chest, abdomen, thighs and pelvis separately. They also consider **extracellular and intracellular water**. A popular brand has 3 series, 6 series and 9 series models, with better features in the higher models.

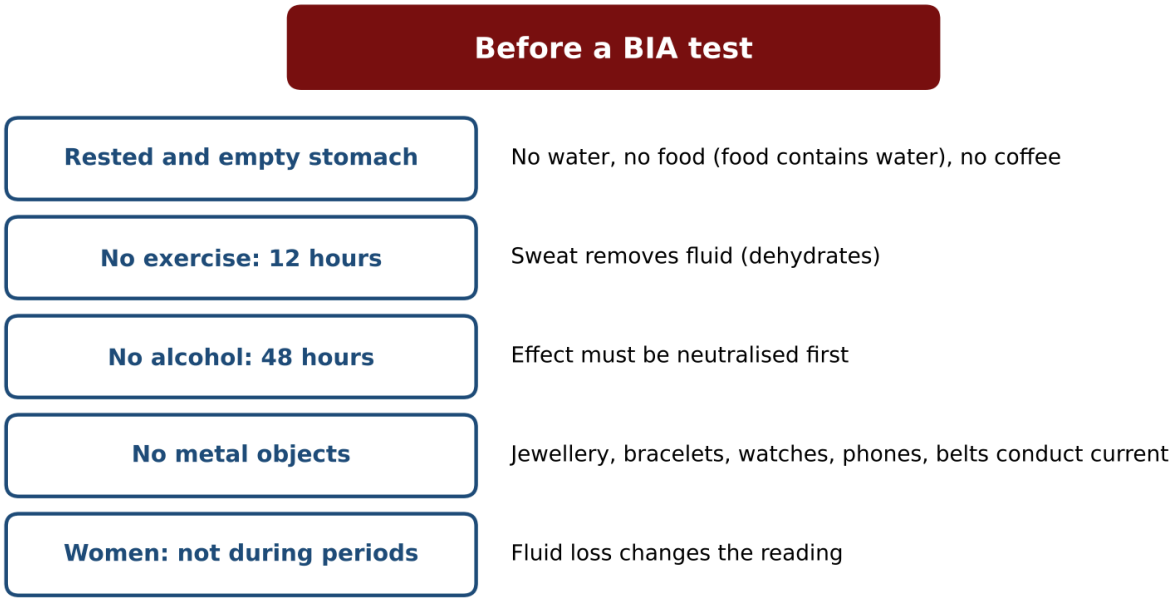
9.4 Demonstration (video)

The video showed the steps:

1. The person steps on the scale barefoot. The footprints are the metal electrodes, and the platform is also a weighing scale, so the machine measures **weight** first.
2. The **height** is entered manually on the screen. The trainer has already measured it with the stadiometer.
3. The person holds the hand electrodes, pressing the **thumbs against the metal part**, and raises the handle slightly to **chest level**, keeping the arms raised and still.
4. The machine runs the test and gives the analysis. Below the weight and height, three results appear: **body fat percentage, muscle mass and BMI**.

9.5 Prerequisites for a BIA test

The conditions must be met before taking the test. The goal is to stand on the machine with **neutral hydration**: nothing extra added from outside and nothing lost.



Prerequisite	Reason
Well rested	Basic requirement
Empty stomach: no water, no food (food contains water)	Water or fluid taken before the test hydrates the person, which is like cheating. The machine does not ignore it: it counts that water and amplifies the current
No coffee	A student asked about black coffee (because getting out of bed is hard). Coffee cannot be had without water, so the answer is no
No alcohol, with a gap of 48 hours	The effect needs to be neutralised before the test
No exercise before the test, with a gap of 12 hours	Sweating loses fluid, so the person is dehydrated and the machine reads the lower hydration level, which is a loss for the person
No metallic objects (ornaments, jewellery, bracelets, watches, gadgets, phones, belts)	Metal conducts electricity, just like water, and can influence the reading
Not during periods (women)	Fluid loss during periods influences the results

Student question: What is the time interval for food (for example 12 hours before)?

A: The preferred condition is an empty stomach along with being well rested. That is why the test is said to be done **first thing in the morning**: there is then automatically a gap of at least about 6 hours since dinner.

9.6 Advantages and disadvantages

Advantages	Disadvantages
Time saving: hardly a minute	**Some models might show standard error.** Not every model is meant, but they cannot be as accurate as the gold standard (so not 100% reliable), though still better than the earlier tests
Easy to use: even front desk staff at a gym can conduct it if the correct method is followed; no certified trainer, physician or exercise physiologist is needed	**Hydration level can impact results:** if the person is dehydrated or over-hydrated, the numbers cannot be relied upon, because the whole principle is that water conducts electricity
Needs only an electric switch and socket: it is not portable like the caliper, but it is small, easy to move, takes little space and needs no heavy equipment	

9.7 Student questions on BIA

Q: Does the machine show the percentage of muscle and fat, or the actual weight in kilograms?

A: It depends on the model. The screen next to the machine is only a display screen connected to a system on which the entire report appears. A very expensive, high-end model shows a **body chart** with the body divided into segments, with **percentages and kilograms** for each segment.

Q: If someone is more hydrated, will the muscle mass show higher?

A: Yes, and that is why the number cannot be relied on in that case. Be careful about hydration.

Q: The price varies from Rs 10,000 to Rs 45 lakh. Does a low-cost machine give more errors, or does a costly one just give more details?

A: Accuracy does not always come with spending. Entry-level or mid-level models may not give features such as in-depth analysis, but the number they give cannot be called totally untrustworthy or completely unreliable.

Using the best test to check a gym's machine. One of the later tests (the final choice, to be revealed in the next lecture) is **not conducted in gyms**, but it gives very accurate, reliable and worldwide accepted numbers. A client can get that test done once, and then compare the gym's BIA machine (entry level or mid level) against it, to get an idea of how close the machine is and how much standard error it has. A student guessed which test this is (DEXA), and the answer was left for the next lecture.

Part 10: The Bod Pod Test (2.3.2)

10.1 Name and method

Bod Pod: **Bod** refers to the **body**, **Pod** refers to a **sealed chamber**. The machine is a big **egg-shaped chamber**. (Similar sleeping pods exist at some airports, where people lie down, a panel

closes, and there is soothing music. The Bod Pod has no such music: the person sits in the machine, the door closes, and the test is run.) It is also called the **air displacement method**.

10.2 Densitometry and the calculation chain

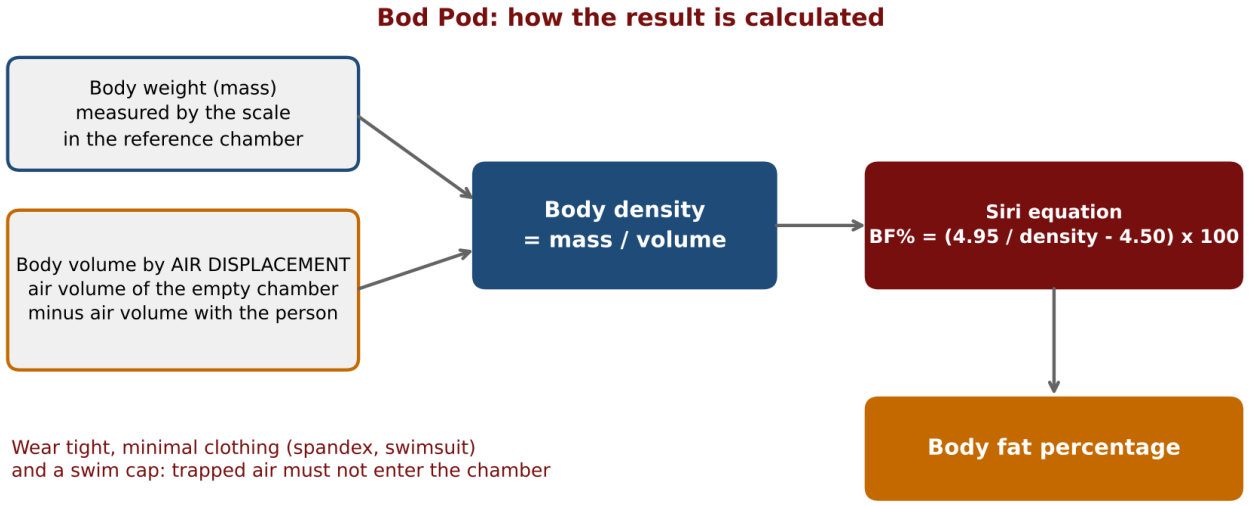
The Bod Pod belongs to the **densitometry methods** of body composition. There are two tests in this group: the Bod Pod, and another test that comes next lecture. Densitometry uses **body density** to measure body fat percentage.

Body density = body mass (body weight) / body volume

Once the machine calculates body density, it applies the **Siri equation** to calculate body fat percentage:

Body fat % = (4.95 / body density - 4.50) x 100

These formulas need not be memorised. The computer does everything, and the technician does not have to do anything. They are explained to show the science behind the machine.



10.3 How the machine works

The slide shows the machine and a cut-away view.

- The part where the person sits is the **test chamber**. Behind the seat is the **reference chamber**.
- In the reference chamber there is the grey box, which is the **weighing scale** that measures body weight.
- A **diaphragm** behind the backrest measures the **volume of air** inside the test chamber.

How body volume is found by air displacement.

1. Before anyone sits inside, the diaphragm measures the **volume of air in the empty chamber** and stores it.
2. The person sits in the chamber. The person takes up some space, so some air is **displaced**. The diaphragm measures the volume of air again, and it is different.

3. The **difference** between the two volumes (empty chamber minus chamber with the person) gives the **volume of the person**.
4. Body mass is already known from the scale, so body density is calculated, and then the Siri equation gives body fat percentage.

Student question: When the door opens, the air volume and pressure change. How is that handled?

A: The machine has already stored the volume of the empty chamber. It analyses both air volume and **air pressure**, and after the door opens, the person sits and the door closes, it compares what the chamber measured earlier with what it measures now. The video also showed a **tube** into which the person breathes out (a small breathing manoeuvre). It measures the **air in the lungs**, so the machine can remove that air as well and consider only the person's body volume. Lung volume measurement improves the accuracy of the test.

Clothing. The person wears a **swimming costume (spandex or swim shorts for men, a swimsuit for women, body-hugging material) and a tight swimming cap**. No extra air must go inside with the person. Ordinary clothes carry air in the shirt and the trouser pockets, and hair has gaps that hold air, so a tight cap is worn.

10.4 Results

In the video the result screen shows body fat percentage and also the masses of fat and fat-free mass. In the example shown, a person's total weight of **97 kg** was made up of **82 kg of fat-free mass** and **15 kg of fat mass**, which can also be expressed as percentages. **Fat-free mass is the same as lean body mass** (confirmed by a student's question). From these figures the fat percentage is 15 / 97, about 15.5% (calculated for these notes; the recording did not state it).

10.5 Advantages and disadvantages

Advantages	Disadvantages
Accurate: comparable to the gold standard. Note that it is **not** the gold standard itself, but accuracy is always compared with the gold standard, so it is almost at par	**Expensive:** not every facility can afford it
Time saving	**Not available in India** (as written on the board); availability is limited
No technical expertise needed: the machine does everything	
Suitable for all ages and special cases: all ages, patients (heart related, joint related, neurological), obese people, children, very elderly people and **hydrophobic** people (people afraid of water). It is safe and easy to use	

Why "hydrophobic" is mentioned. The next test, the water displacement method (hydrostatic or underwater weighing), measures body volume with the person **submerged in a tub of water**. A person afraid of water, a patient, an obese person, a child or a senior citizen may not manage to go underwater and stay there during the test. The Bod Pod, in which the person only sits in a chamber, is safe for them.

The slides add: pretty accurate compared with hydrostatic weighing, easy to operate, short measurement time, well suited to all populations (children, obese, elderly, disabled); disadvantages: expensive and not easily available.

Two key parameters for the final choice. When all tests are compared to pick the best one, **affordability and availability** will also be considered, along with accuracy. There is no point in a very accurate test that is not available in the country, or that is too expensive for the facility or the client.

Part 11: Wrap-up and Preview of Fitness Testing 2

11.1 Summary of this lecture

Topic	Key points
Objectives	Global: standardisation, performance benchmarking. Client: baseline, progress tracking, goal setting, strengths and areas to work on
Pre-step	Health screening with PAR-Q (all No, otherwise NOC) and informed consent
Order	BP, body composition, cardiorespiratory endurance, muscular strength, muscular endurance, flexibility
BP	Systolic (upper, 90 to 120) and diastolic (lower, 60 to 80); seven-step procedure; classification chart
Anthropometric	WC (visceral fat risk), WHR (overall risk), BMI (category); no body fat percentage
Skinfold	Caliper, subcutaneous fat, plus or minus 3%
BIA	Water conducts current; muscle 75% water, fat 10%; prerequisites
Bod Pod	Air displacement, body density, Siri equation

11.2 Comparison of the body composition tests so far

Test	Gives % body fat?	Measures	Accuracy	Main advantages	Main disadvantages
WC	No	Circumference (risk from visceral fat)	Risk only	Simple, no machine	No percentage
WHR	No	Circumference ratio (overall risk)	Risk only	Simple, no machine	No percentage
BMI	No	Weight and height (category)	Low reliability	Easy formula	Cannot tell fat from muscle, misses skinny fat
Skinfold caliper	Yes	Subcutaneous fat	Plus or minus 3%	Portable, easy, quick	No visceral fat, not for very obese or very lean, pinching
BIA	Yes	Resistance to current (visceral and subcutaneous)	Some models have standard error	Quick, easy, needs only a socket	Hydration sensitive
Bod Pod	Yes	Body volume by air displacement	Comparable to gold standard	Accurate, quick, suits all populations	Expensive, availability

11.3 What comes next

Two machines remain for body composition: the **hydrostatic (water displacement) test** and the second remaining machine (the lecture used the numbers 2.3.3 and 2.3.4). Once those are covered in **Fitness Testing 2**, the best test for body composition will be disclosed (after comparing pros and cons, including cost and availability), and the lectures will move on to the other components.

Points to Confirm

These are places where the recording was unclear or inconsistent, or where it differs from the slides or E-book. The lecture's version is used in the notes.

1. **BMI worked example numbers.** The recording converts 5 ft 8 in to 68 in correctly (the speaker briefly said "5 ft 6" but the board shows 5 ft = 60 in and 68 in). It then multiplies 68 by 0.0254 and says the result is 1.7526, but 68 × 0.0254 = **1.7272 m** (1.7526 is the value for 69 in). The board rounds to 1.7 m, squares it to 2.9 (1.7 × 1.7 = 2.89) and gets BMI **31.03** (board 31.02 and 31.03). The slides use 1.72 m, 2.95 and **30.50**. With the exact 1.7272 m the BMI is about **30.17**. All versions fall in the obese category. These notes show the lecture's calculation and note the exact value.
2. **BMI categories.** The lecture and slides use the proposed **Asian criteria** (underweight below 18.5, normal 18.5 to 22.9, overweight 23 to 24.9, obese 25 and above). The lecture said

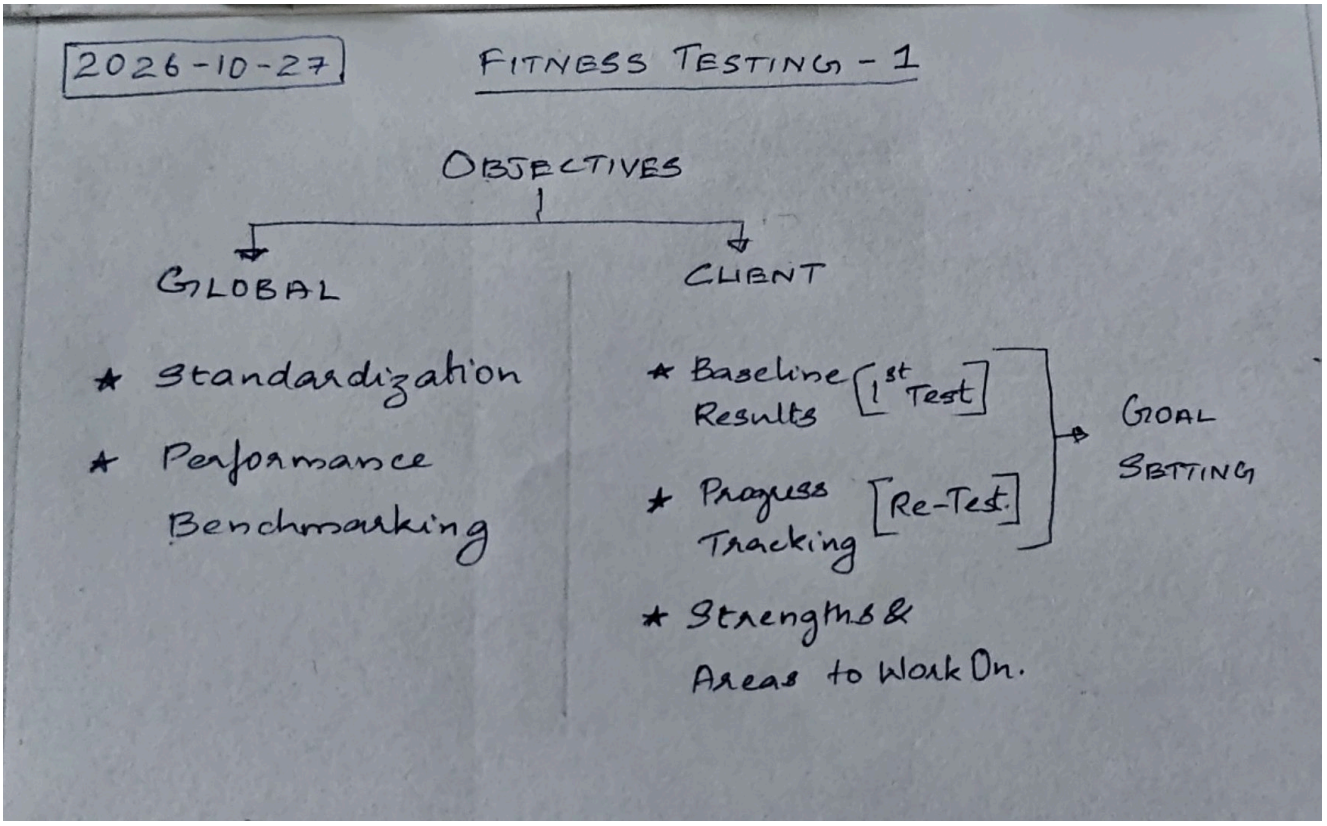
"anything more than 25 is obese", and the slide prints "> 25". The E-book (Chapter VII) uses overweight 25 to 29.9 and obese 30 and above. The recording did not give the overweight band, which is taken from the slide.

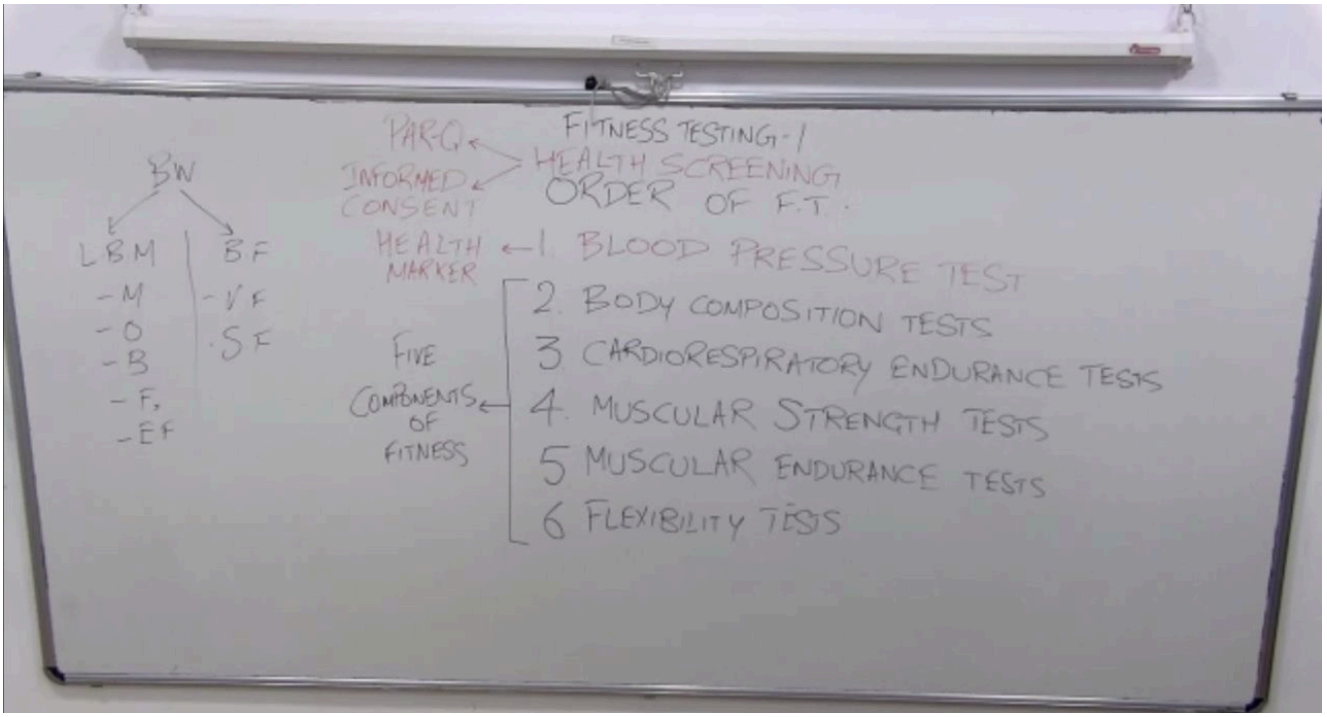
3. **Waist circumference thresholds.** The lecture and slides give **90 cm (35.5 in) for men and 80 cm (31.5 in) for women** for Indians and Asians. The E-book's obesity risk factor table uses waist above 102 cm for men and 88 cm for women (and BMI above 30, WHR above 0.95 for men and 0.86 for women). 35.5 in is 90.2 cm, so 90 cm is an approximation.
4. **WC measuring site wording.** The lecture says across the iliac crest, passing through the belly button (the widest part). The slides say the greatest anterior extension of the abdomen, usually at the level of the umbilicus and directly above the iliac crest. The E-book abdomen site is at the greatest anterior extension, usually at the level of the umbilicus. For WHR the lecture and slides agree (midpoint of the last rib and the iliac crest), and the E-book describes the waist as the narrowest part of the torso.
5. **WHR case study.** The lecture used a male (low risk) and then a female (high risk) with the same numbers; the slide shows only the female case (high risk). Both are consistent. The E-book says WHR risk limits vary with age and sex (for example 0.95 for young men and 0.86 for young women, with higher limits at 60 to 69).
6. **Skinfold sites.** The lecture and slides agree: chest (midpoint between nipple and top of the armpit), abdomen 1 inch from the belly button and mid thigh for men; triceps, suprailiac and mid thigh for women. The E-book describes an abdomen site 2 cm to the right of the umbilicus, a chest fold at one-half the distance between the anterior axillary line and the nipple, and says measurements are taken on the right side of the body.
7. **What the caliper error is compared with.** The lecture says plus or minus 3% compared with the gold standard (without naming it); the slide says compared with underwater weighing; the E-book also gives plus or minus 3%.
8. **Which test is the "gold standard".** The lecture did not name it (a student guessed DEXA and the answer was left for the next lecture). The slides and E-book describe hydrostatic (underwater) weighing as the gold standard in percent body fat measurement. Confirm in Fitness Testing 2.
9. **PAR-Q length.** The lecture said 10 or 12 questions; the slide sample has 10 questions.
10. **Blood pressure classification boundaries.** The slide table has overlapping boundaries (for example 120 appears in both normal and elevated, and 80 as the top of normal diastolic and the start of stage 1). The lecture only stated the normal range (90 to 120, 60 to 80), the low BP rule (systolic below 90 or diastolic below 60) and the stage 2 rule (140 or higher, or 90 or higher). Elevated, stage 1 and crisis values are taken from the slide.
11. **Three-day blood pressure rule.** Repeating the check over three days when BP is high is the lecture's practice and is not in the slides. The E-book instead says to take at least 3 readings and use the average (and to measure supine and standing positions).
12. **Order of testing.** The lecture and slides list six steps (blood pressure, body composition, cardiorespiratory endurance, muscular strength, muscular endurance, flexibility). The E-book lists five (resting measurements such as heart rate and blood pressure, body composition, cardiorespiratory fitness, strength, flexibility) and does not list muscular endurance separately.
13. **BIA prerequisites.** The lecture lists empty stomach (no food, water or coffee), no exercise (12 hours) and no alcohol (48 hours). The E-book says to avoid eating and drinking within 4 hours of the test, avoid exercise within 12 hours, avoid alcohol within 48 hours, void the

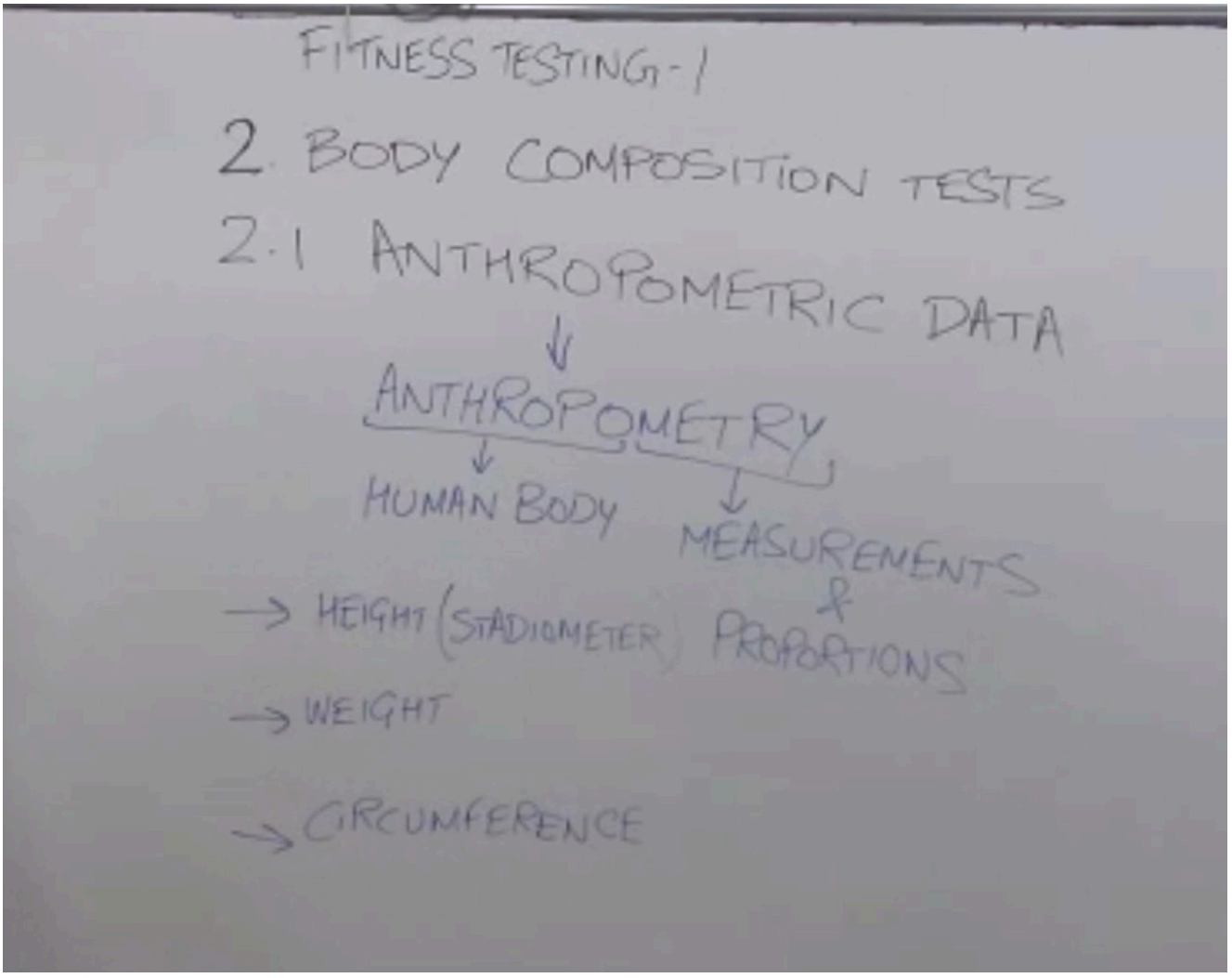
bladder completely and avoid diuretics. The "at least 6 hours since dinner" figure was given in answer to a student question.

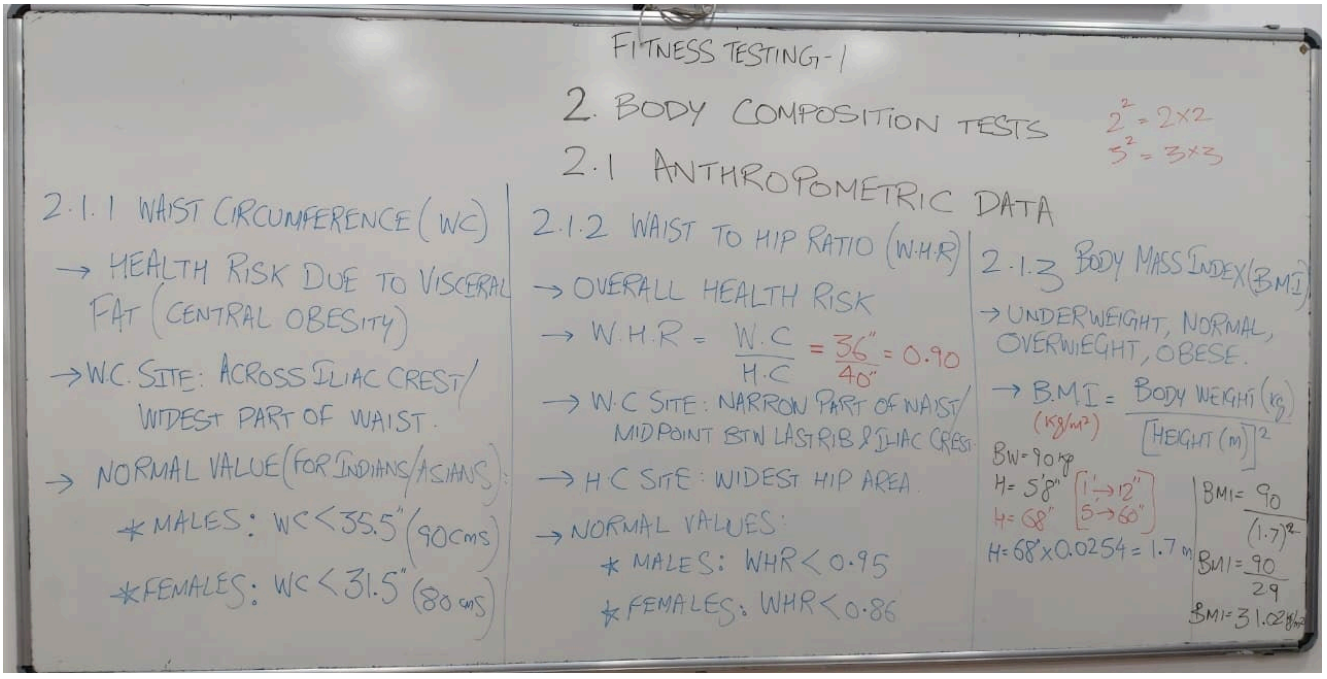
14. **BIA accuracy.** The lecture says some models show standard error and that BIA is better than the previous tests. The E-book gives about plus or minus 3% when done correctly, and says BIA tends to overestimate lean people and underestimate obese people.
15. **Bod Pod.** The lecture calls it time saving, comparable to the gold standard but not the gold standard, expensive, with "not available in India" written on the board. The E-book gives plus or minus 3% error, a validity of r = .93 against hydrostatic weighing, a measurement time of about 5 to 8 minutes, a cost of about 30,000 to 40,000 US dollars and says only a few facilities have it.
16. **Prices quoted in the lecture.** Caliper about Rs 300 to Rs 5,000; BIA machines Rs 10,000 up to Rs 45 lakh. These are approximate, spoken figures.
17. **Water and composition percentages.** Muscle 75% water, 20% protein, 4% IMTG and 1% glycogen; fat 10% water and 90% triglyceride. These come from the earlier body composition lecture and are not on the slides or in the E-book (the slides only say lean mass has more water and salts).
18. **Brand name unclear.** The brand of high-end BIA machines with 3, 6 and 9 series models was not clear in the recording and is not named here.
19. **Numbering of the remaining machine tests.** The lecture numbered BIA 2.3.1 and Bod Pod 2.3.2 and said the other densitometry test would be 2.3.3. The slides list the machines in the order BIA, hydrostatic weighing, Bod Pod, DEXA. The numbers for the last two are therefore not confirmed.
20. **Bod Pod example.** The recording gave 97 kg, 82 kg fat-free mass and 15 kg fat mass but did not state the percentage; 15.5% is calculated in these notes. The Siri equation was spoken as "4.95 divided by body density minus 4.50 into 100" and is written here as (4.95 / density minus 4.50) x 100, which matches the E-book form 495 / density minus 450.
21. **Ideal body fat levels.** Recalled in the lecture as less than 15% for males and less than 20% for females, and matched by the slide.
22. **Board photos.** One board photo (the blood pressure example: 115/75, systolic, diastolic and normal ranges) could not be cleaned properly, so its content is shown as a diagram instead. Photos that only showed slides are rebuilt as tables.

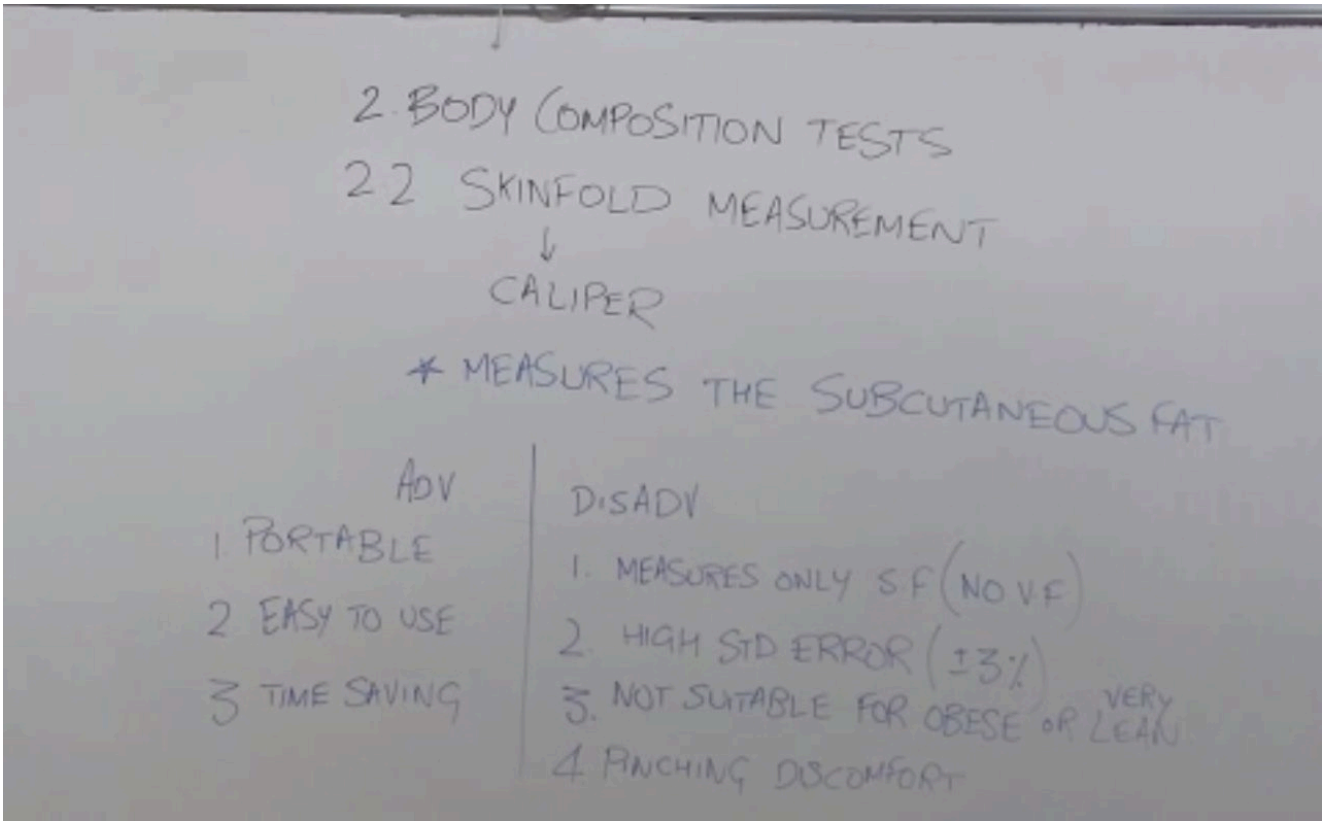
Board Photos from the Lecture

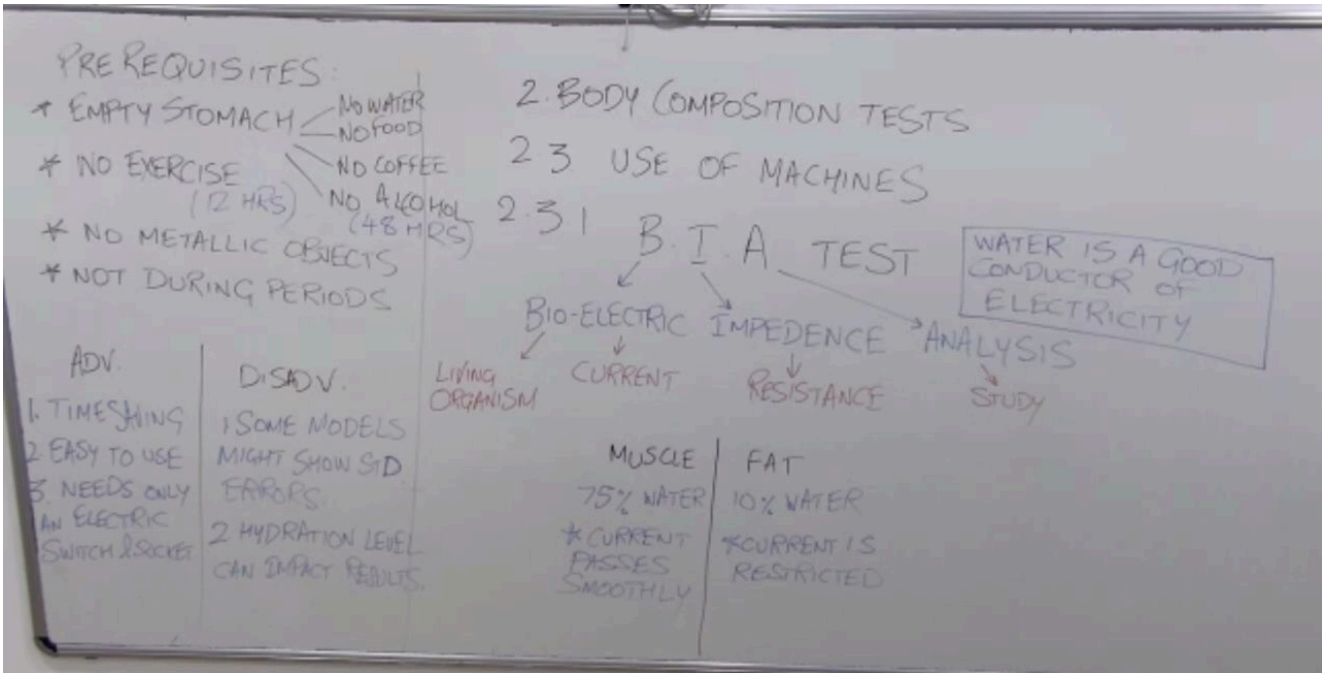


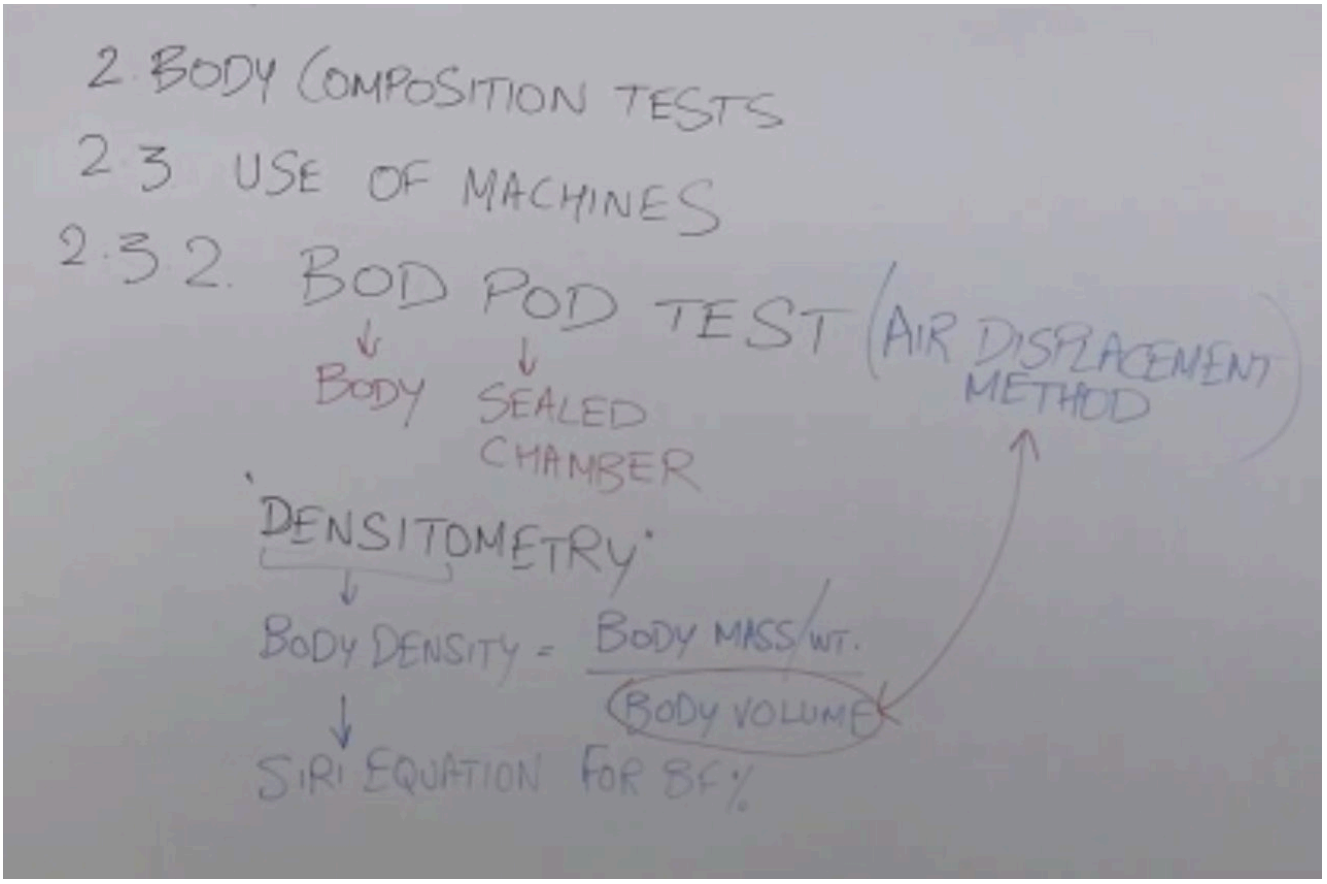


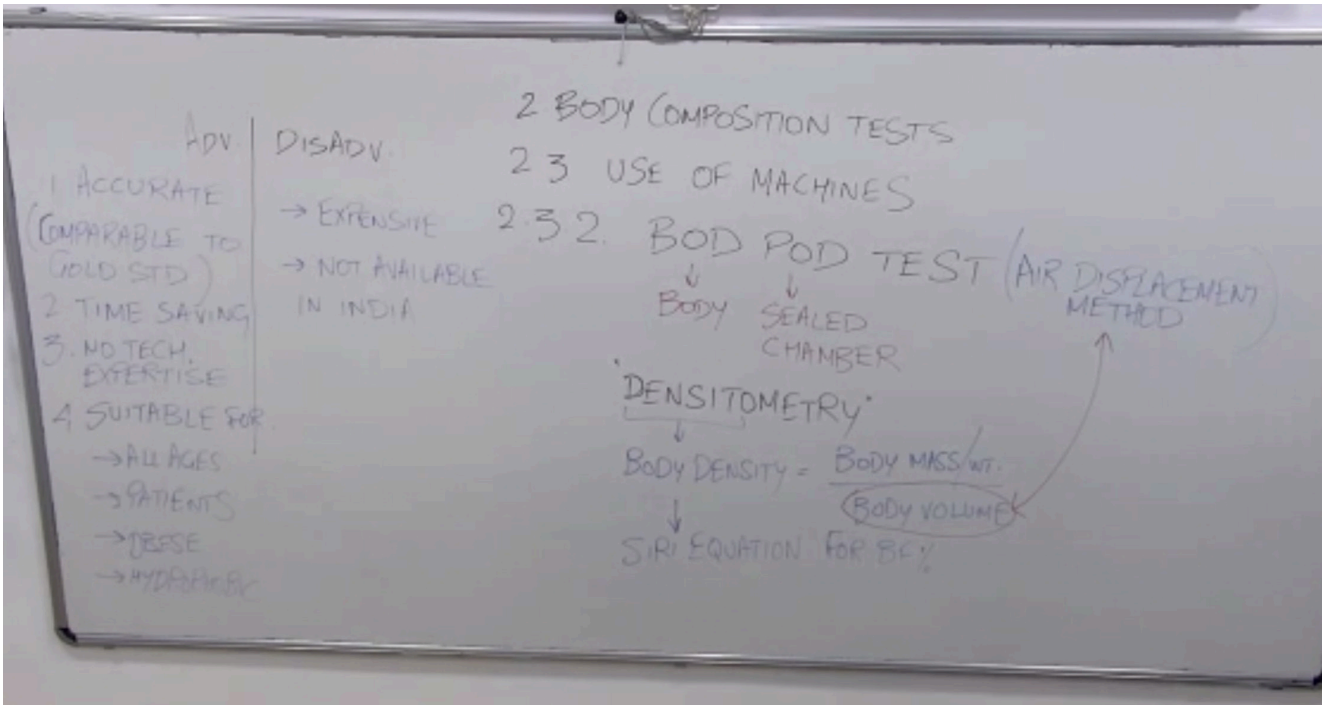

FITNESS TESTING-1
2. BODY COMPOSITION TESTS
2.1 ANTHROPOMETRIC DATA
ANTHROPOMETRY
HUMAN BODY
MEASUREMENTS & PROPORTIONS
→ HEIGHT (STADIOMETER)
→ WEIGHT
→ CIRCUMFERENCE


FITNESS TESTING-1
2. BODY COMPOSITION TESTS
2.1 ANTHROPOMETRIC DATA
2.1.1 WAIST CIRCUMFERENCE (WC)
→ HEALTH RISK DUE TO VISCERAL FAT (CENTRAL OBESITY)
→ W.C. SITE: ACROSS ILIAC CREST/ WIDEST PART OF WAIST.
→ NORMAL VALUE (FOR INDIANS/ASIANS):
* MALES: WC < 35.5" (90cms)
* FEMALES: WC < 31.5" (80 cms)
2.1.2 WAIST TO HIP RATIO (W:H:R)
→ OVERALL HEALTH RISK
→ W.H.R = W.C / H.C = 36"/40" = 0.90
→ W.C SITE: NARROW PART OF WAIST/ MIDPOINT BTW LAST RIB & ILIAC CREST
→ H.C SITE: WIDEST HIP AREA
→ NORMAL VALUES:
* MALES: WHR < 0.95
* FEMALES: WHR < 0.86
2.1.3 BODY MASS INDEX (BMI)
→ UNDERWEIGHT, NORMAL, OVERWEIGHT, OBESE.
→ B.M.I = BODY WEIGHT (kg) / [HEIGHT (m)]^2 (kg/m^2)
BW = 90 kg
H = 5'8" [1' → 12", 5 → 60"]
H = 68
H = 68 × 0.0254 = 1.7 m
BMI = 90 / (1.7)^2
BMI = 90/29
BMI = 31.02 kg/m^2
2^2 = 2×2
3^2 = 3×3


2. BODY COMPOSITION TESTS
2.2 SKINFOLD MEASUREMENT
↓
CALIPER
* MEASURES THE SUBCUTANEOUS FAT
ADV
1. PORTABLE
2. EASY TO USE
3. TIME SAVING
DISADV
1. MEASURES ONLY S.F (NO V.F)
2. HIGH STD ERROR (±3%)
3. NOT SUITABLE FOR OBESE OR VERY LEAN
4. PINCHING DISCOMFORT


PREREQUISITES:
* EMPTY STOMACH — NO WATER NO FOOD
* NO EXERCISE (12 HRS) — NO COFFEE, NO ALCOHOL (48 HRS)
* NO METALLIC OBJECTS
* NOT DURING PERIODS
2. BODY COMPOSITION TESTS
2.3 USE OF MACHINES
2.3.1 B.I.A TEST
WATER IS A GOOD CONDUCTOR OF ELECTRICITY
BIO-ELECTRIC IMPEDENCE ANALYSIS
LIVING ORGANISM
CURRENT
RESISTANCE
STUDY
ADV.
1. TIMESAVING
2. EASY TO USE
3. NEEDS ONLY AN ELECTRIC SWITCH & SOCKET
DISADV.
1 SOME MODELS MIGHT SHOW STD ERRORS.
2 HYDRATION LEVEL CAN IMPACT RESULTS.
MUSCLE
75% WATER
* CURRENT PASSES SMOOTHLY
FAT
10% WATER
*CURRENT IS RESTRICTED


2 BODY COMPOSITION TESTS
2.3 USE OF MACHINES
2.3.2. BOD POD TEST (AIR DISPLACEMENT METHOD)
BODY
SEALED CHAMBER
DENSITOMETRY
BODY DENSITY = BODY MASS/WT.
BODY VOLUME
SIRI EQUATION FOR BF%


ADV.
1. ACCURATE (COMPARABLE TO GOLD STD)
2. TIME SAVING
3. NO TECH. EXERTISE
4. SUITABLE FOR
→ALL AGES
→PATIENTS
→OBESE
DISADV.
→ EXPENSIVE
→ NOT AVAILABLE IN INDIA
2 BODY COMPOSITION TESTS
2.3 USE OF MACHINES
2.3.2. BOD POD TEST (AIR DISPLACEMENT METHOD)
BODY
SEALED CHAMBER
DENSITOMETRY
BODY DENSITY = BODY MASS/WT.
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